Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - I

Expansive Soils: Problems, Identification, Swell Pressure, Improvement and Under-reamed Piles

Syllabus topics covered in this unit

- Problems of Expansive Soils
- Tests for Identification of Expansive Soils
- Methods of Determination of Swell Pressure
- Improvement of Expansive Soils
- Foundation Techniques in Expansive Soils
- Under-reamed Piles

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Expansive Soils - Introduction

Expansive soils are fine grained clayey soils that show large *volume change* - they *swell* on absorbing water and *shrink* on losing water. The movement is not caused by load but by change in moisture content. They are found in arid and semi-arid regions where evaporation is greater than rainfall.
In India these are the *Black Cotton (BC) soils* (regur), covering about *20 percent* of the land - Deccan plateau, Maharashtra, Madhya Pradesh, Gujarat, Karnataka and *Andhra Pradesh* (*Rayalaseema, Kurnool, Anantapur, Guntur, Krishna*). They are rich in lime, iron and magnesia, and dark in colour because of titaniferous magnetite and humus.

Why do they swell? - clay mineralogy

Clay mineral	Structure	Swelling	Specific surface (m^2/g)
Montmorillonite (smectite)	2:1 layer (silica-alumina-silica), weak oxygen-oxygen bond	Very high; water enters between the layers	800
Illite	2:1 layer with potassium bond	Medium	80
Kaolinite	1:1 layer with hydrogen bond	Low	15

- *Montmorillonite* is the main cause: its sheets are bonded weakly so water and exchangeable cations enter between them and push the sheets apart (*interlayer swelling*).
- *Diffuse double layer* and *osmotic* repulsion adsorb more water when the electrolyte concentration is low.
- Soil particle is flat and negatively charged; it attracts cations (Na^+, Ca^{2+}) and dipolar water. Sodium montmorillonite swells more than calcium montmorillonite.

Factors affecting swelling

- *Type and amount of clay mineral* (montmorillonite %), cation exchange capacity.
- *Initial water content* - dry soil swells more; *dry density* - denser soil swells more.
- *Soil structure* - flocculated vs dispersed; *compaction* method (static / dynamic).
- *Surcharge* - higher load reduces swelling; *drainage and climate* - wet-dry cycle, vegetation (trees remove moisture), leaking pipes, water table fluctuation.
- *Time* - swelling is time dependent, mostly in the first monsoon seasons after construction.

2. Problems of Expansive Soils

The seasonal moisture change in the top active zone (depth 1.5 to 3.5 m in India, up to 5 m or more in arid areas) causes cyclic heave and shrinkage and repeated movement of foundations and structures. The depth of the active zone is the depth up to which seasonal moisture variation takes place.

Effects on structures

Structure	Typical damage
Light buildings (1 - 2 storeys), compound walls	Diagonal and vertical cracks in walls, door and window frames jammed, floors cracked and heaved, differential settlement and tilting; "doming" in the centre and edge shrinkage in dry period
Pavements, runways, canal lining	Longitudinal cracking, waves and undulations, edge failure, uplift and breaking of lining, seepage loss
Retaining walls, basements	Lateral swelling pressure larger than active earth pressure; tilting, cracking
Pipelines, culverts, service lines	Breakage, leakage at joints which then wets the soil further
Earth dams, embankments, cuttings	Slope failure (shallow slides), cracks from desiccation
Deep foundations - piles	Uplift (jacking) of the pile by adhesion in the active zone, loss of contact
Transmission towers, tanks	Tilting, uplift of footings

- Crack pattern: in the dry period cracks start at the corners and wide at the top (edge shrinkage); in the wet period the centre heaves (centre lift) - cracks wide at the bottom.
- Greatest damage to lightly loaded structures because the swelling pressure (50 - 500 kPa and more) is greater than the structural load.
- Movement is non-uniform - near the walls the soil is shielded from evaporation and stays moister than the soil in the open, so edge and centre differ.
- Vegetation (trees) near buildings dries the soil locally and causes shrinkage settlement.
- The annual loss in India due to damage of structures on expansive soil runs to thousands of crores of rupees.

Remember

* Swell = heave on wetting; shrinkage = settlement and cracks on drying; the cycle repeats every year.
* Expansive soil is not weak (high strength when dry) - the problem is volume change, not bearing capacity.

3. Tests for Identification of Expansive Soils

Expansive soils are identified by field indications, index property tests and direct swell tests. Mineralogical tests give the clay type.

Field identification

- Dark grey / black colour, deep wide cracks (polygonal pattern) in summer, sticky and plastic in the monsoon, soap-like feel.
- Local history of building damage; presence of calcareous nodules (kankar) in lower layers.

Laboratory tests

Test	Method (IS)	Remarks
Free swell index (FSI)	IS 2720 Part 40: 10 g oven dried soil passing 425 micron into two 100 ml graduated jars - one with water, one with kerosene; read volume after 24 h	$FSI = (V_d - V_k) / V_k \times 100$ %
Atterberg limits	Liquid limit (LL), plastic limit (PL), shrinkage limit (SL), plasticity index PI = LL - PL	High LL and PI -> high swell; low SL
Colloid content / clay size fraction	Hydrometer analysis, % finer than 0.002 mm	Higher clay -> more swelling
Activity (Skempton)	A = PI / (% clay fraction < 2 micron)	A > 1.25 active; montmorillonite A = 1 - 7
Linear shrinkage	Shrinkage of bar in a mould	Used with LL
Cation exchange capacity (CEC)	Ammonium acetate method	Montmorillonite: 80 - 150 meq/100 g
Mineralogical analysis	X-ray diffraction (XRD), DTA, electron microscope, SEM	Direct identification of the mineral

Free swell index (IS 1498 / IS 2720-40)

- $FSI\ (\%) = (V_d - V_k) / V_k \times 100$; V_d = volume in distilled water, V_k = volume in kerosene (non polar, no swelling).
- Example: V_d = 18 cc, V_k = 10 cc => FSI = 80 percent (high degree of expansiveness).

Classification by degree of expansion (IS 1498)

Free swell index (%)	Degree of expansion	Probable damage
< 20	Low	Negligible
20 - 35	Moderate	Some damage
35 - 50	High	Considerable
> 50	Very high	Severe

Classification by index properties (Holtz and Gibbs / Chen)

LL (%)	PI (%)	SL (%)	Colloid (%)	Probable expansion (% - dry soil at 7 kPa)	Degree
20 - 35	< 18	> 15	< 15	< 10	Low
35 - 50	15 - 28	10 - 15	13 - 23	10 - 20	Medium
50 - 70	25 - 41	7 - 12	20 - 31	20 - 30	High
70 - 90	> 35	< 11	> 28	> 30	Very high

- Chen (1988) by PI: 0 - 15 low; 10 - 35 medium; 20 - 55 high; > 35 very high potential for swell.
- Seed, Woodward and Lundgren (1962) relate swell potential S to activity and clay fraction: $S = 3.6 \times 10^{-5} A^{2.44} C^{3.44}$ (C = clay %, A = activity).
- Differential free swell (DFS) of the soil is also used by Indian engineers: DFS < 20 = low and > 50 = very high.
- Direct test: one dimensional oedometer swell test - swell % = change in height / original height x 100 under the seating load.

4. Determination of Swell Pressure

Swell pressure is the pressure that must be applied to prevent any volume increase of a soil sample when it is given free access to water. Swelling pressure is the maximum pressure developed (kPa). It tells the structural load needed or the uplift force on a footing or pile.

Methods (IS 2720 Part 41 - oedometer tests)

Method	Procedure	Remarks
1. Constant volume method	Undisturbed (or remoulded) sample in the oedometer ring. Add water; as the sample tries to swell, add load to keep the dial gauge reading constant (no change in height). The final load / area = swell pressure	Direct and quick; the sample is not allowed to swell at all; tends to give the lower value for high swelling soil
2. Swell-consolidation (loading after swell) method	Let the sample swell freely under seating load, then load it in the usual increments until it is compressed back to its original height. The pressure at the original void ratio e_0 = swell pressure	Gives higher value; the soil is disturbed by swelling; takes longer
3. Free-swell / Zero swell (Method of 3 or 4 specimens)	Several identical specimens are wetted under different surcharges (e.g., 6.25, 12.5, 25, 50 kPa); measure the % swell / heave in each and plot against log pressure; pressure for zero swell = swell pressure	Needs more samples; gives the swell-load relation; reliable
4. Double oedometer / Jennings & Knight	One sample at natural moisture and another soaked are loaded; compare the e - log p curves	Also gives collapse and heave prediction

Constant volume method - steps

Trim sample, fit ring, saturate porous stones → Seat load 5 kPa; add water → Add load to keep height constant → Max load at equilibrium = P_s

Empirical relations

* Komornik and David (1969): $\log P_s$ (kg/cm^2) = 2.132 + 0.0208 LL + 0.000665 gamma_d - 0.0269 w
* Typical swell pressure: LL 50 -> 50 - 100 kPa; LL 80 -> 300 - 600 kPa; BC soil in India: 50 - 400 kPa.
* Heave of a layer: Delta H = (C_s H / (1 + e_0)) log ((sigma_0 + sigma_s) / sigma_0) (C_s = swell index)

- Other methods: swell pressure from the swell-load curve (Fredlund), Free swell + consolidation (Sridharan), Chen's correlation with PI, and in field plate load test with water.
- Use: the dead load pressure should be more than P_s to avoid heave; if not, adopt foundation techniques or improve soil. Swell pressure acts on retaining walls and basement walls as extra lateral load.

5. Improvement of Expansive Soils

The aim of improvement is to reduce the swell potential, or to control the moisture variation, or to remove the soil from the active zone. The methods are of four groups.

A. Replacement and moisture control

- Replacement - remove expansive soil up to the active zone depth and replace it with a non-expansive granular soil. Costly when the depth is large.
- Cohesive Non-Swelling (CNS) layer (IS 9451 / Katti): a layer of 0.9 to 1.2 m (min 0.6) of non-swelling cohesive soil (PI < 20, low swell) placed below foundations, canal lining and pavements. The CNS layer gives overburden and a cushion, reduces suction, and so swell is controlled to the lower depth.
- Pre-wetting (ponding) - soil is allowed to swell before construction by ponding water for weeks; heave is completed beforehand. Slow, works for large flat sites (roads, canals).
- Moisture control - keep the moisture constant: footpath / apron (min 1.5 m wide) around the building, impermeable horizontal and vertical moisture barriers (polyethylene, bitumen, concrete), good surface drainage, plumbing leaks avoided, no trees close to the building (distance > height of the tree), avoid flower beds near the walls.
- Compaction control - compact on the wet side of OMC with low dry density (e.g., 95 percent of the standard Proctor dry density at OMC + 2 percent) to reduce swell.

B. Chemical stabilisation

Stabiliser	Action	Dose
Lime	Cation exchange (Ca^{2+} replaces Na^{+}), flocculation, pozzolanic reaction forms cementing gel -> lower PI, swell and plasticity	4 - 8 percent by weight (initial consumption of lime, ICL by pH test - Eades Grim)
Cement	Hydration cements the particles; for low PI soils	5 - 10 percent
Fly ash + lime	Pozzolanic reaction; reduces cost and swell	Lime 3-5 %, fly ash 10-20 %
Calcium chloride, sodium chloride (salt)	Reduce the double layer, hygroscopic - reduce moisture loss	1 - 3 percent
Organic cationic compounds, potassium compounds, enzymes (EKM), Terrazyme	Reduce the water affinity of clay	Manufacturer dose
Bitumen emulsion	Waterproofing the soil	Special cases

- Lime piles / lime columns: boreholes (dia 0.2 - 0.4 m) filled with quicklime or lime-sand; lime diffuses into the soil. Lime slurry pressure injection (LSPI) through boreholes up to 3 - 4 m depth; the slurry penetrates the cracks of the soil.
- Lime treatment produces a quick reduction in plasticity, but also needs curing; sulphate-bearing soils may heave on lime treatment (ettringite) - check.

C. Mechanical and physical methods

- Surcharge loading (placing an overburden to counter the swelling), dynamic compaction, thermal treatment (heating to 400 C decreases the swell by changing the clay), soil reinforcement with fibres or geosynthetics, geocell mattress, remove / sand cushion under footings.
- Prewetting + lime / CNS combination is used under canals (e.g., Nagarjuna Sagar canal, Tungabhadra).

D. Structural and foundation approach

Instead of changing the soil, the structure may be isolated from the movement - see the next section on foundation techniques.

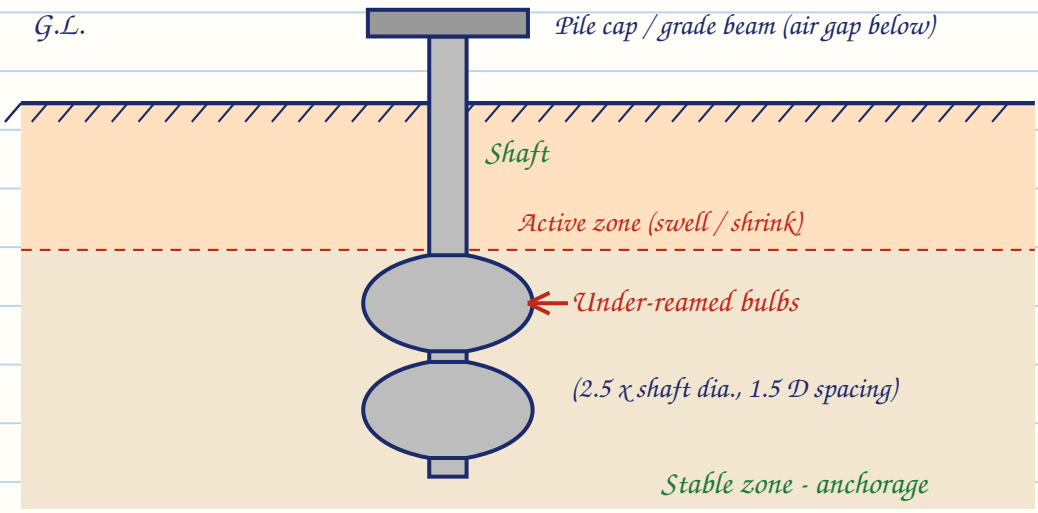
6. Foundation Techniques in Expansive Soils

The foundation is placed so that it is either below the active zone, or isolated from the soil movement, or is rigid enough to resist the movement. Common choices are the following.

Technique	Details
Deep foundation below the active zone	Drilled piers / bored piles / under-reamed piles anchored in the stable zone below the seasonal moisture change; resist uplift by their anchorage
Foundation on CNS layer / sand cushion	Strip or isolated footings on a 0.9 - 1.2 m CNS layer; shallow foundations at 1.0 - 1.5 m (but need caution); swell pressure on footing is reduced
Raft foundation / stiffened raft	Rigid RC raft or waffle raft with stiffening beams to move as one unit without bending (the structure "floats" on the moving soil); for light buildings
Isolated footing with tie beams	Footings on a sand cushion + plinth / tie beams (RC bands) at plinth, lintel and roof levels to give rigidity
Belled piers / drilled shafts	Large diameter drilled piers with bell; with void form below the grade beam
Pier and beam with void space	Grade beams spanning between piles, with a void (air gap 8 - 10 cm) below the beam so the swelling soil does not push the beam
Shallow foundation with controlled moisture	Moisture barrier around building, vertical barrier to depth of 1.5 m, apron, drain; reduce moisture changes
Structural design measures	Light weight structure, flexible joints, expansion joints and separate sections, slip joints at doors, plumbing with flexible connections, no brick work in direct contact (slip layer between plinth masonry and the beam)

- Principle: (i) Avoid the swelling zone; (ii) Increase the dead load more than the swell pressure; (iii) Make the structure rigid / flexible to match the movement; (iv) Prevent the water from reaching the soil.
- Tie beams and lintel bands in buildings. Roof level sloped for drainage, fix the plinth protection up to 1 m and no cut in the plinth by pipes.
- Retaining walls: use a granular backfill, drainage, design for swell pressure.
- Pavement and canal: CNS layer, lime treatment, prewetting, sub-surface drain, narrow shoulder with impermeable cover.

7. Under-reamed Piles



Under-reamed pile is a bored cast-in-situ concrete pile with one or more bulbs (under-reams) formed by enlarging the bore at the lower part. It was first developed in India by Central Building Research Institute (CBRI), Roorkee, for black cotton soils. IS 2911 (Part 3) covers the design and construction.

Why under-reamed piles in expansive soils?

- The pile is taken below the active zone, in the stable moisture zone, where the soil does not swell or shrink.
- The bulbs act as anchors against the uplift forces caused by the swelling soil acting on the shaft in the active zone (the soil tries to lift the shaft by skin friction).
- The bulbs also give additional bearing capacity (end bearing and annular bearing).
- Suited for light and medium loaded structures; can be built with simple hand operated equipment.

Geometry (IS 2911 Part 3)

Feature	Recommended value
Shaft diameter	20 - 50 cm (20, 25, 30, 35, 40 cm commonly)
Bulb diameter D_u	2.0 to 3.0 times the shaft diameter (usually 2.5 d for black cotton soil, 2.0 d in hard soil); 2.5 d is most used
Number of bulbs	Single or double (multi) bulb; double under-reamed piles are common
Spacing of bulbs	Not more than $1.5 \times D_u$ centre to centre; the top bulb at not less than $2\ D_u$ below the ground (usually below the active zone)
Minimum length	3.5 m below the ground level in expansive soil (not less than 3 m) and the bulbs below the active zone depth
Spacing of piles	Not less than $2 \times D_u$ (for under-reamed) and 3 d for straight shaft
Reinforcement	Minimum 0.4 % longitudinal steel (min 3 bars, 10 mm) + stirrups 6 mm at 15 cm; for uplift 0.4 to 0.8 %
Concrete	M 15 to M 20; grade beam 20 cm min

Construction steps

Bore the shaft with auger (hand / rig) → Under-ream with the cutter tool (bulbs) → Place the reinforcement cage → Pour concrete by tremie; grade beam over the pile

- Boring dry: no bentonite for BC soil; casing in soft soils; the under-reaming tool has two hinged blades that open by pushing the tool down.
- Concreting immediately after boring to prevent collapse; tremie / hopper to avoid segregation; compact by rodding.
- Grade beam (plinth beam) connects the pile heads and supports the wall; keep an air gap of 8 - 10 cm (sand / void form) between the beam bottom and the ground so that the swelling soil does not lift the beam.
- Pile top is anchored into the grade beam. Piles also under compound walls and in canal bridges.

Load carrying capacity (IS 2911 Part 3)

Ultimate load of single pile in clay

- $Q_u = A_p N_c c_p + A_u N_c c_u$ + alpha $c_s A_s$ (compression)
- A_p = base area of the bottom bulb; A_u = annular area of the other bulbs $(pi/4)(D_u^2 - d^2)$; $N_c = 9$; c_p, c_u = cohesion at bulb levels; alpha = adhesion factor (0.3 - 0.5 for BC soil, often shaft friction in the active zone is neglected); A_s = shaft surface area below the active zone.
- Safe load $Q_s = Q_u / FS$, $FS = 2.5$ (compression), and for uplift $Q_a = Q_u / 3$ in uplift.
- Under-reamed pile capacity: 25 to 80 kN (single bulb 20-30 cm shaft), 100 - 250 kN for bigger piles.

Design for uplift (swelling soil)

- Uplift force $F_u = pi\ d\ L_a\ tau_s$ where L_a = depth of the active zone and tau_s = swelling soil shear (about 0.5 c or 20 - 50 kPa).
- Check: anchorage resistance in the stable zone $Q_{anch} = c N_c A$ (bulbs) + side friction >= F_u - W (dead load) with a factor of safety about 1.0 - 1.5 on the uplift; if not, increase the length, add bulbs or increase the load by dead weight.
- Steel for the tension at the pile - usually the area $A_{st} = F_u / (0.87 f_y)$ with the minimum percentage.
- Pile length should be at least 3 m and extend above the equilibrium moisture depth; many designs take the length $L = 3.5$ to 5 m with two bulbs.

Advantages

- Simple technique, low cost, uses local labour; suitable for BC soils and also for loose granular and filled up soils; resists uplift and loads, little settlement.
- Limitations: not for soft clays with high water table (bulb collapse), special tools needed; the bulbs cannot be properly formed in loose sand below the water table.

Quick revision of Unit I

- Expansive soil: montmorillonite; BC soil; swell-shrink cycle; active zone 1.5 - 3.5 m.
- Identification: FSI, DFS, LL, PI, SL, activity; IS classification of FSI (< 20 to > 50).
- Swell pressure: constant volume, swell-consolidation, zero swell methods. Improvement: CNS layer, lime, prewetting, moisture barrier, replacement.
- Foundations: under-reamed piles (bulb = 2.5 d, spacing 1.5 D_u, min length 3.5 m, below active zone), raft, grade beams with air gap.

Important Questions (JNTUA pattern)

- Explain the mechanism of swelling in expansive soils. List the factors affecting swelling.
- Describe the problems caused by expansive soils to buildings, pavements and canals.
- Explain the tests used for identification of expansive soils. How is the free swell index determined?
- Explain the methods of determining the swelling pressure of a soil with a neat sketch.
- Discuss the methods of improvement of expansive soils. Explain the CNS layer concept and lime stabilisation.
- What are the foundation techniques used in expansive soils? Explain the isolation techniques.
- Explain under-reamed piles: geometry, construction and design. Why are they suitable in black cotton soil?

Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - II

Dewatering and Grouting

Syllabus topics covered in this unit

- Methods of De-watering
- Sumps and Interceptor Ditches
- Single and Multi Stage Well Points, Vacuum Well Points
- Horizontal Wells, Foundation Drains, Blanket Drains
- Criteria for Selection of Fill Material Around Drains
- Electro-osmosis
- Objectives of Grouting, Grouts and Their Properties
- Grouting Methods: Ascending, Descending and Stage Grouting
- Hydraulic Fracturing in Soils and Rocks
- Post Grout Tests

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1. Dewatering - Introduction and Methods

Dewatering is the removal of *ground water or surface water* from a construction site (excavation, trench, foundation, tunnel) to keep the work area dry and the sides stable. It is also used to lower the water table permanently (drains) to improve the soil strength.

Need for dewatering

- Excavation below the water table is unsafe - *sides collapse*, bottom heaves, *quick condition* (piping) develops when the upward seepage gradient $i = i_c = (G - 1)/(1 + e)$ about 1.
- Dry condition is needed for foundation work, concreting and compaction; reduce the *lateral pressure* on the excavation supports; increase the effective stress and thus the strength; avoid *buoyancy* of structures.

Classification of methods

Group	Methods	Suitable soil
1. Open sump / ditch (surface) methods	Sumps, interceptor ditches, pumping from open pits	Coarse soils: gravel, coarse sand, rock; clean low flow
2. Well point systems	Single stage, multi stage, vacuum, eductor, deep well	Sand, silty sand (k about 10^{-2} to 10^{-5} cm/s)
3. Deep wells	Deep well with submersible pump, ejector wells	Large drawdown > 15 m, permeable soils
4. Horizontal wells / drains	Horizontal drains, foundation drains, blanket drains	Slopes, dams, long excavations
5. Electro-osmosis	Direct current across soil	Silts, clays ($k < 10^{-5}$ cm/s)
6. Cut-off and exclusion methods	Sheet piles, diaphragm wall, slurry trench, grouting, freezing	Where pumping is not allowed or inflow is large

The choice depends on *permeability (grain size)*, depth of the draw down, area to be dewatered, duration, cost, effect on neighbours. A general guide is: gravel - sump; sand - well points; fine sand and silt - vacuum well point; clay - electro-osmosis.

Darcy and flow to a well

Basic formulae

- Darcy: $q = k\,i\,A$. Steady flow to a fully penetrating well in an unconfined aquifer (Dupuit): $Q = \pi k (H^2 - h^2) / \ln (R/r)$
- Confined aquifer (thickness b): $Q = 2\pi k b (H - h) / \ln (R/r)$
- R = radius of influence (Sichardt): $R = 3000\, s \sqrt{k}$ (s = drawdown in m, k in m/s, R in m)
- For a long trench / line of wells: $q = k (H^2 - h^2) / (2L)$ per unit length.

2. Sumps and Interceptor Ditches

Sump pumping (open pumping)

A sump is a pit dug lower than the excavation level in a corner or along the side, at which water seeps in by gravity and is pumped out by a centrifugal / submersible pump. It is the simplest and cheapest method.

- Sump size: about 1 m x 1 m or more, depth 1 - 2 m below the formation level; lined with perforated drum / precast rings / brick with gravel filter to prevent fines from entering.
- Sump pit at the lowest corner; a perimeter drain leads water to the sump. Pump capacity > the inflow.
- Advantages: simple, cheap, small plant, suited to gravel and coarse sand and rock excavation with small depth (< 1.5 - 2 m below water table).
- Limitations: in fine sands the upward seepage causes piping / quick sand, boiling at the bottom, sloughing of slopes, unstable; depth limited by suction lift (about 6 - 7 m max).

Interceptor (cut-off) ditches

- Interceptor ditch (catch drain) is a ditch dug around the upslope side of the excavation or slope to intercept the surface runoff and shallow ground water before it reaches the works. The water is led by gravity to a natural drain or to a sump.
- Section: trapezoidal 0.5 - 1.0 m wide; the invert slope 1 in 200 or more; sometimes lined with concrete or geotextile and filled with gravel (French drain) with perforated pipe.
- Used for road cuttings, hill-side slopes, embankment toe, dam abutments, landslide stabilisation; lowering the water table in the slope improves stability.
- Interceptor drains are also built as a trench perpendicular to the flow of the groundwater, in which a perforated pipe in filter material collects the water.

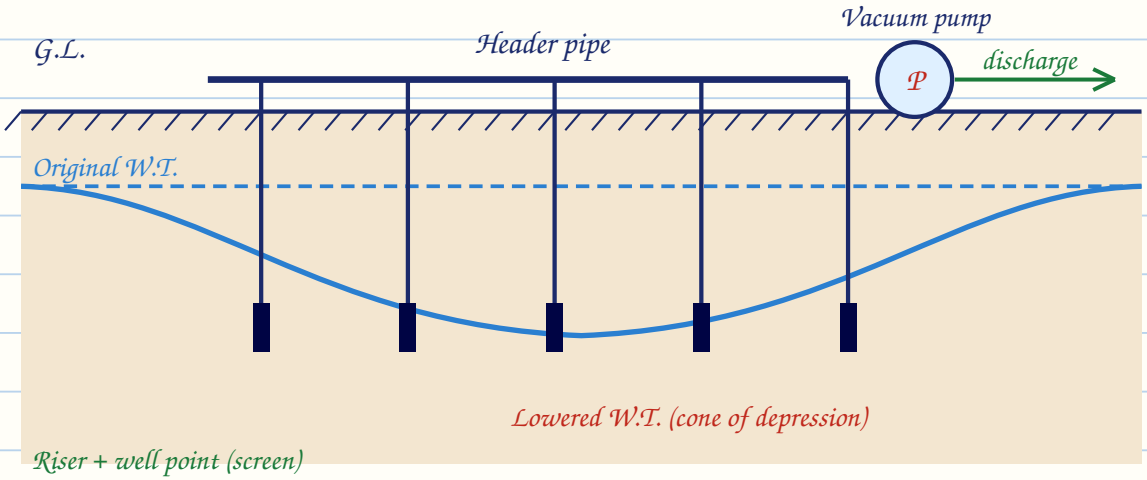
Surface water and seepage → Interceptor ditch (collect) → Sump → Pump and discharge

Remember

- Sump and ditch methods = open pumping, lowest cost, but not for fine soils or deep excavation.
- Filter around the sump pit is a must; slope the excavation floor towards the sump.

3. Well Point Systems

A well point is a small diameter (40 - 50 mm) perforated tube (screen about 0.5 - 1 m long and covered with a mesh) driven or jetted into the ground. A line of well points spaced 0.75 - 3 m apart (usually 1 - 2 m) is connected to a common header pipe (150 - 200 mm) and a centrifugal / vacuum pump that draws the water by suction.



Components

- Well point (screen + tip with jetting valve), riser pipe (40 - 50 mm), swing connection (flexible hose), header pipe, pump (vacuum assisted self-priming) and the discharge pipe.
- Installation: by jetting with water pressure (6 - 8 kg/cm^2) into the sand, giving a filter sand column of coarse sand around the screen, in a hole of 20 - 25 cm.

Single stage well points

- One line (or ring) of well points around the excavation; the water table is lowered up to about 5 - 6 m below the pump level (limit of the suction lift).

- The header at about the ground level. Spacing: 1 m in fine sand, 2 m in coarse sand; use closer spacing at the more permeable parts. A double sided system (both sides of the trench) for wide excavation > 6 m wide.
- Pump capacity: yield of one well point 0.1 - 1 l/s; a pump of 150 mm handles 100 - 150 well points.

Multi stage well points

- For depth of the draw down greater than about 5 - 6 m: well point lines are set at two or more levels (stages) one above the other in a stepped excavation (benches) - the upper stage is installed first and excavated, then the next stage lower.
- Each stage lowers the water table by 4 - 5 m. Needs a wide excavation (benches 1.5 m or more wide, side slopes), more pumps, more cost, large water volume. Alternative: deep wells / ejector well points if the depth > 12 m.

Vacuum well points

- In silty and fine sands ($k < 10^{-3}$ cm/s), gravity flow is slow, so a vacuum (about 0.6 - 0.8 atmosphere) is maintained in the well point and sand filter (by sealing the top of the filter with clay/ bentonite). The vacuum increases the hydraulic gradient and the soil water is drawn out; it also consolidates the soil (additional effective stress).
- The filter sand column up to the ground surface is sealed. Spacing is closer (0.75 - 1.5 m). Used for deep trenches in fine silty soils, stabilising slopes, tunnels in silty sand.
- Ejector (eductor) wells use high pressure water jet (ejector) at the bottom - for depth up to 30 m at low yield in fine soils.

Aspect	Single stage	Multi stage	Vacuum
Max drawdown	About 5 - 6 m	10 - 15 m (2 or 3 stages)	Increases gradient; about 6 m per stage
Soil	Sand, coarse silt	Sand	Silty sand, fine silt
Spacing	1 - 3 m	1 - 2 m per stage	0.75 - 1.5 m
Cost	Low	High	Medium - High

Design of a well point system (steps)

Soil data and k → Required drawdown, flow Q → No. of well points = Q / q_{wp} → Pump and header selection

- Total flow to the system is found by treating the system as a large well of equivalent radius r_e = sqrt(A / pi) (A = area enclosed): $Q = \pi k (H^2 - h^2) / \ln (R / r_e)$ for an unconfined aquifer.
- The drawdown at the centre is s, and extra 0.5 - 1 m below the formation level is provided. Well point yield $q_{wp} = 2 \pi r_s L_s$ (sqrt(k) / 15) (Kozeny / Sichardt).

4. Horizontal Wells, Foundation Drains and Blanket Drains

Horizontal wells / horizontal drains

- Horizontal (or slightly upward inclined) perforated pipes (50 - 100 mm) drilled or pushed into a slope or a dam, 30 - 100 m long with 2 - 5 % upward inclination, to drain the ground water from the slope face by gravity; the end is connected to a drain pipe. Ideal for landslides, cuttings, dam abutments.
- Spacing 3 - 15 m; the length extends beyond the potential slip surface. Pipes are slotted PVC wrapped with geotextile. Advantages: no pumping, low maintenance, lowers pore water pressure (r_u) and increases the factor of safety.

Foundation drains

- Foundation drain (perimeter drain / toe drain) is a trench around the foundation of a building or a retaining wall, below the footing level, with a perforated pipe (100 - 150 mm) in gravel envelope wrapped in filter fabric and sloping at 1 in 100 to a sump or storm drain.
- Purpose: remove the water around footings and basement, reduce the hydrostatic uplift and the lateral pressure on the walls, prevent the dampness and the softening and swelling of the soil.
- Under-drains below the floor slab (gravel + pipe) in basements and also behind the retaining wall (weep holes 75 mm at 3 m, with the gravel pocket).

Blanket drains

- Blanket (horizontal) drain is a layer of coarse granular material (sand and gravel 0.3 - 1.0 m thick, filter graded) laid under an embankment, earth dam, road, or a structure, to collect the seepage and discharge it to the toe drain.
- In earth dams it is the horizontal drain at the downstream third: keeps the phreatic line within the section, protects the downstream slope from the seepage and piping; the chimney drain is the vertical equivalent.
- Under road embankments it is a sub-base drainage layer (granular sub-base drainage layer, geotextile separator) to remove the water from the pavement and prevent the pumping and frost.
- Under buildings on expansive soil / floors: sand blanket (0.15 - 0.3 m) with perforated pipes.

Seepage water → Filter layer → Blanket / pipe → Toe drain or sump

Design of blanket drains (concept)

- Capacity: $q = k_d\, i\, t$ (k_d = permeability of the drain material, t = thickness).
- The drain permeability should be at least 100 times that of the soil being drained; adequate thickness; free from clogging by the filter layer.

5. Criteria for Selection of Fill Material Around Drains

The filter (envelope) material around a drain must (i) retain the soil particles so that no fines are washed in (piping) and (ii) allow water to pass freely (permeable), and (iii) not clog with time. These requirements are expressed by the Terzaghi filter criteria, extended by Bertram and the US Army Corps (USBR).

Filter criteria (Terzaghi - Bertram)

- Piping (retention) criterion: D_{15} (filter) / D_{85} (soil) < 5
- Also D_{50} (filter) / D_{50} (soil) < 25 and for uniform soils $D_{15f} / D_{85s} < 5$
- Permeability (drainage) criterion: D_{15} (filter) / D_{15} (soil) > 5
- Uniformity: filter gradation curve nearly parallel to that of the soil; coefficient of uniformity $C_u = D_{60}/D_{10} < 20$ for the filter.
- Fines: the filter should have less than 5 percent finer than 75 micron (No. 200 sieve), and no plastic fines.

- Pipe slot / hole criterion: D_{85} (filter) > 1.2 x the slot width or > 1.0 - 2 x the hole diameter (so that the filter does not enter the pipe).
- When the drain / filter is two stage, apply the criteria to each successive layer (soil -> fine filter -> coarse filter -> pipe).
- The filter must be durable (hard, round or sub-angular, no soft particles), not prone to chemical clogging, easy to compact without segregation.
- Geotextile filter may replace the sand filter: apparent opening size (AOS) $O_{95} < 2 - 3\, D_{85}$ of the soil; permittivity higher than needed; adequate clogging resistance (gradient ratio test).

Soil to drain	Typical filter / fill material
Clay / silt	Coarse sand (0.1 - 5 mm), concrete sand; geotextile wrapped gravel
Fine sand	Medium sand / fine gravel in 2 layers
Coarse sand and gravel	Gravel 5 - 20 mm; stone, pea gravel
Collection pipe	Gravel 10 - 40 mm with D_{85} larger than slots

Example

- Soil: D_{85} = 0.2 mm, D_{15} = 0.02 mm. Then the filter D_{15} must be < 5 x 0.2 = 1.0 mm and > 5 x 0.02 = 0.1 mm => choose D_{15} about 0.3 - 0.8 mm (medium sand).

6. Electro-osmosis

Electro-osmosis is the movement of pore water through a fine grained soil (silt, clay) from the anode (+) towards the cathode (-) when a direct current is passed through the soil. Used when the soil is too fine ($k < 10^{-5}$ cm/s) for the gravity or vacuum drainage; first used by Casagrande (1939) in Germany.

- Principle: clay particles are negatively charged and the water has a diffuse layer of positive ions (cations). Under the electric field the cations move to the cathode, dragging the water with them. The water collects at the cathode (a perforated well point) and is pumped away.
- Electrodes: anode - steel rod / pipe (metal, corrodes) or any, cathode - perforated well point (steel tube). Spacing 3 - 6 m (anode-cathode), voltage 40 - 180 V (gradient of 50 V/m), DC power from a generator.

Quantity of flow

- $q = k_e i_e A$ (k_e = coefficient of electro-osmotic permeability, i_e = voltage gradient (V/m), A = area)
- k_e is about 5 x 10^{-5} cm/s per V/cm and is almost independent of the grain size - so it is the same for all soil types (the reason it works well in clays: for clay $k_{hydraulic}$ about 10^{-8} but k_e about 5 x 10^{-5}).
- Example: i_e = 0.5 V/cm => electro-osmotic flow velocity v = 5 x 10^{-5} x 0.5 = 2.5 x 10^{-5} cm/s (about 2 cm/day).

Applications and effects

- Dewatering and stabilising silts and soft clays in deep excavations, slopes (stabilising of landslides, Norwegian clay), consolidation of soft clay, pile driving (reduces the friction, increases the strength after the current: electro-osmotic piles).
- Drainage of tunnels, increase in the shear strength by consolidation and by the chemical effect (electro-chemical hardening with injection of $CaCl_2$ or Al salts at the anode).
- Reduce the water content, increases the strength by 2 - 3 times. Also remediation of contaminated clays (electrokinetic).

Advantages	Limitations
Works in clays and silts where other methods fail	High power cost (energy about 20 - 200 kWh/m^3)
Quick effect on the strength; no effect on the nearby structures	Anode corrosion; the soil near the anode dries and cracks, resistance increases
Can be combined with the well points and sheet piles	Not suitable in salty soil (high conductivity - wasted current) or with buried metal

7. Grouting - Objectives and Grouts

Grouting is the process of injecting a fluid material (grout) under pressure into the voids, cracks or fissures of soil or rock, where it hardens (sets) and thereby improves the strength and reduces permeability of the ground.

Objectives (uses) of grouting

- Reduce permeability - seepage control (curtain grouting below dams, cut-off), tunnels, shafts, excavations.
- Increase strength and bearing capacity - strengthen weak soil and rock; underpin old foundations.
- Reduce settlement, fill voids (cavities, mine workings, sinkholes), compaction grouting.
- Stabilise loose sands against liquefaction, excavation stability, support of tunnels, seal leaking walls and rock.
- Fill the gap behind tunnel linings, pre-stress and anchor (rock bolt grouting), repair the concrete (crack repair), raising of slabs (slab jacking).

Grouts - types and properties

Grout	Composition	Use / remarks
Cement grout (suspension)	Cement + water (w/c 0.4 - 2.0), + bentonite 1 - 3 %, fly ash, admixtures	Rock fissures, coarse gravel and sand; cheap; particles > 0.1 mm - cannot enter fine sand (voids < 3 x D_{85})
Micro-fine cement	Cement < 10 micron (d_{95} < 15 micron), slag	Fine sand, narrow fissures
Clay (bentonite) grout	Bentonite + water; + cement	Cheap, low strength; seepage cut-off; reduces cement bleeding
Chemical: sodium silicate (water glass)	Sodium silicate + reagent ($CaCl_2$, formamide, bicarbonate) forms silica gel	Medium to fine sand; Joosten two shot, one shot Siroc; low viscosity; strength 0.5 - 3 MPa
Chemical: acrylamide, polyurethane, resins (epoxy)	Polymer grouts	Very fine sand; leak sealing; costly, toxicity (acrylamide banned in many countries)
Bituminous grout	Asphalt emulsion	Seepage sealing in rock
Lime / lime-fly ash, mortar grout	Lime slurry, sand-cement mortar	Large voids, cavities, mines; fill (compaction grout = low slump mortar)

Properties of the grouts

- Viscosity (resistance to flow - low viscosity enters finer pores), fluidity; Penetrability (the ability to enter the soil - groutability ratio $N = D_{15}$ (soil) / D_{85} (grout) > 25, and D_{10} (soil) / D_{95} (grout) > 11 for suspension grout).
- Setting time (gel time) - must be long enough to place and short enough to avoid the dilution; bleeding (water separation, < 2 - 5 % in 2 h), stability (no segregation), strength, durability, permeability after setting, shrinkage, compatibility with the ground water chemistry (sulphates), toxicity and cost.
- Rheology: Newtonian (chemical) and Bingham fluid (cement, clay - yield stress). Pressure and the rate of injection control the spread.
- Groutability: ratio of grain size of the soil to that of the grout. Particulate grout: soil permeability k > 10^{-1} cm/s; chemical grout k > 10^{-3} cm/s.

8. Grouting Methods - Ascending, Descending, Stage

In rock and in deep soil the grouting is done in sections (stages) of the hole length - usually 3 to 6 m each - using packers to isolate the section under treatment. The different sequences are called the

grouting methods.

1. Ascending (upstage) method

- The hole is drilled to the full depth at once. Grouting is begun from the bottom section and then the packer is raised to the next section above. (also called "packer method up-stage").
- Packer set above the section, grout is injected under the pressure, then raised to grout the next, upwards.
- Advantages: quick, drilling is done once, cheap; Disadvantages: the grout may leak (by-pass) along the hole and up, fissures upstream may be left ungrouted, pressure at the upper zone limited, not suitable for the badly fractured rock, the hole may collapse.

2. Descending (downstage) method

- The hole is drilled for the first stage (top section), grouted, the grout is allowed to set, the hole is re-drilled (washed) through the set grout and deeper by the next stage, and so on downwards.
- Advantages: the upper zone already sealed so that higher pressures may be used in the lower stages; best control; grouting of the poor rock is effective; less lifting and surface leaks; used for dam foundations, fractured rock and tunnels.
- Disadvantages: slow and costly (the waiting time for the setting, re-drilling).

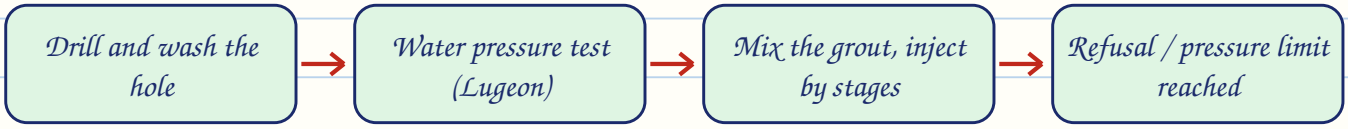
3. Stage grouting (Stage by stage / packer)

- The hole is divided into stages; each stage is grouted with a packer (single or double) and the sequence depends on the rock: stage grouting can be ascending stage or descending stage; the packer type is a single packer or double (straddle) packer to isolate a single zone.
- The stage length 3 - 6 m (3 m for poor rock, 6 m for good rock).
- Tube-a-manchette (sleeve pipe) method for soils: a plastic pipe with rubber sleeves at 0.33 - 1 m spacing is grouted into the bore with a sheath grout, then a double packer injects the grout through any sleeve at the needed pressure, multiple times - very controlled in alluvium and sand.

Aspect	Ascending	Descending
Drilling	Full depth once	Stage by stage, re-drilling
Time and cost	Less	More
Control and quality	Lower	Higher
Pressure	Limited by leakage	Higher in the lower stages
Best for	Good, sound rock; soil	Weak, broken and fissured rock, dams

Sequence of the holes (split spacing)

- Primary holes at 12 m spacing; secondary at 6 m; tertiary at 3 m (split spacing); check holes at the end. The grout take decreases in each series - the sign of a successful treatment.
- Curtain grouting (1 - 3 rows), blanket / consolidation grouting (shallow, closely spaced to the grid of 3 - 6 m), contact grouting (between the concrete and rock), jet grouting (high pressure 20 - 40 MPa jet cuts the soil and mixes the cement to create columns - soil-crete), compaction grouting (stiff mortar bulbs densify the soil), permeation grouting.



- Pressure limit: about 1 kPa per cm depth, i.e., $p = 0.23 - 0.25\ kg/cm^2$ per m depth of the overburden (about 22 kPa/m); to avoid heave and hydraulic fracture. Refusal when the take is < 1 l/min at the max pressure for 10 min.
- Grout mix changed: start thin (w/c = 5 : 1) and thicken (3:1, 2:1, 1:1, 0.8:1) as the take continues.

9. Hydraulic Fracturing in Soils and Rocks

Hydraulic fracturing (hydrofracture) is the splitting of the ground by the fluid pressure when the injection pressure exceeds the minimum principal stress plus the tensile strength of the soil / rock. The fracture is a new crack opened by the fluid, which is then filled by the grout - forming lenses and sheets.

Mechanism

- In a borehole the fluid pressure p_f is raised; around the hole the tangential (hoop) stress becomes tensile. When p_f reaches the breakdown pressure the soil fractures.
- Fracture occurs perpendicular to the minimum principal stress - in the field generally vertical fractures at the shallow depth where horizontal stress ($sigma_3$) is the minimum, and horizontal fractures where the vertical stress is the minimum (below about 8 - 10 m in the normally consolidated soil it is horizontal when the $K_0 > 1$).
- Breakdown pressure: $p_b = sigma_3 + T_0 - u$ (fracture in rock); in soil $p_b = 2\ sigma_3 - u + c$ (approx. from the cavity theory) - the typical injection pressure ratio $p_f / sigma_v$ about 1.2 - 2.0.

Uses

- Deliberate hydrofracture: to grout fine grained soils (clay, silt) that cannot be permeated - a network of grout sheets / lenses compresses the soil, raises the structure (compensation grouting

under the tunnel and buildings), soil improvement of clay, and to increase the permeability of the rock for oil and geothermal wells and groundwater wells (yield).

- Measurement of in-situ stress of rock (hydraulic fracturing test) - shut in pressure = sigma $_3$.
- Unwanted hydrofracture in dams: the cracks in the core of the earth dam (hydraulic fracturing of the core - a cause of dam piping), and in the grouting at too high pressure causing the heave or uplift of the ground and rock mass, opening the existing joints, and leaks.

In soils	In rocks
Fracture by the tension around the hole; grout lenses in the weakest plane; fine soils are strengthened by the grout lenses and consolidation	Opens the existing joints / bedding planes; the pressure limited by overburden (rock cover) to avoid the lifting
Pressure monitored by the pressure-time and flow-time curves - a sudden drop in pressure with an increase in flow indicates fracture	Pressure increased in steps; Lugeon test to check the permeability before grouting
Used in sleeve pipe in clay and compensation grouting	Used for the curtain below the dam, for gas, oil and geothermal wells

10. Post Grout Tests

After the grouting, quality control tests verify that the planned improvement (strength, permeability, continuity) has been achieved. They are done after the grout has cured (7 - 28 days).

Test	Purpose / procedure
Water pressure (Lugeon) test	Packer test in the check holes: water is injected at 10 kg/cm^2 for 10 min; 1 Lugeon = 1 litre/min per metre of the hole at 10 kg/cm^2; target < 1 - 3 Lugeon after the grouting
Permeability / pumping test	Insitu permeability with falling or constant head test in boreholes or pumping from wells
Core drilling and inspection	Check holes - cores show the grout in the fissures, Rock Quality Designation (RQD) improved
Unconfined compression test on the cores / samples	Strength of grouted soil; strength gain ratio
Standard penetration (SPT), cone penetration (CPT), pressure-meter	Compare the penetration resistance before and after (e.g., in sands N increases from 8 to 25)
Plate load test	Bearing capacity and the modulus after the grouting
Geophysical tests	Seismic cross-hole / down-hole velocity increases; electrical resistivity; ground penetrating radar
Excavation (pit) / visual	Directly see the grouted mass
Flow meter / piezometers	Seepage and the pore pressure under the dam compared with the design
Grout take records	Decreasing take from primary to tertiary holes proves the closure

- Check holes are drilled at 5 - 10 % of the total holes at the locations of the likely weakness (between the grout holes).
- The acceptance criteria are stated: (i) permeability less than the specified (say 1 x 10^{-5} cm/s, or 3 Lugeon), (ii) minimum strength, (iii) no loss of the water in the check holes.

Quick revision of Unit II

- Dewatering: sump (gravel, coarse sand), well points (sand; single < 6 m, multi stage deeper, vacuum for silty sand), horizontal drains, blanket drains, electro-osmosis for clay (k_e = 5 x 10^{-5} cm/s per V/cm).
- Filter: $D_{15f}/D_{85s} < 5$ and $D_{15f}/D_{15s} > 5$.
- Grout: cement, clay, chemical (silicate); groutability ratio; ascending / descending / stage (packer); hydrofracture when $p > sigma_3 + T$; post grout: Lugeon test, cores, SPT.

Important Questions (JNTUA pattern)

- Explain the methods of dewatering and the criteria for their selection with respect to the soil type.
- Write about sumps and interceptor ditches. What are their limitations?
- Explain the single stage, multi stage and vacuum well point systems with neat sketches.
- Explain the horizontal wells, foundation drains and blanket drains.
- What are the criteria for the selection of the filter material around drains? Give an example.
- Explain the principle and application of electro-osmosis. Derive the flow equation.
- What are the objectives of grouting? Explain the types of grouts and their properties.
- Explain ascending, descending and stage grouting methods and compare them.
- Write short notes on hydraulic fracturing in soils and rocks and post grout tests.

Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - III

Densification Methods in Granular and Cohesive Soils

Syllabus topics covered in this unit

- In-situ Densification in Granular Soils: Vibration at Ground Surface, Impact at Ground Surface
- Vibration at Depth, Impact at Depth
- In-situ Densification in Cohesive Soils: Preloading or Dewatering
- Vertical Drains: Sand Drains, Sand Wick, Geodrains
- Stone and Lime Columns
- Thermal Methods

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Need for Densification and Classification of Methods

Densification (compaction in place) increases the density of loose soil, which increases shear strength and bearing capacity, reduces settlement and compressibility, permeability, and the danger of liquefaction. The method depends on the soil type.

- Granular soils (sand, gravel; free draining): densified by rearranging the grains into a denser state by vibration or impact - the soil volume reduces immediately.
- Cohesive soils (clay, silt; low permeability): the density is raised only by squeezing the water out (consolidation) over time, so they are treated by preloading, drains, columns, and heat.

Soil	Principle	Methods
Granular (cohesionless)	Particle rearrangement by vibration / impact; relative density D_r increases	Surface: vibratory roller, plate vibrator, dynamic compaction (drop weight at the surface). Depth: vibro-compaction (vibroflotation), compaction piles, blasting, deep dynamic compaction
Cohesive (fine grained)	Expulsion of water (consolidation), replacement or reinforcement, chemical change	Preloading and surcharge, vertical drains (sand drain, wick drain), stone columns, lime columns, electro-osmosis, thermal methods

Relative density and compaction

- $D_r = (e_{max} - e) / (e_{max} - e_{min}) \times 100$ percent; D_r < 15 very loose, 15 - 35 loose, 35 - 65 medium, 65 - 85 dense, > 85 very dense.
- Targets: raise D_r to 70 - 85 percent (SPT N > 20 - 30) for foundations on sand and to avoid liquefaction.

Depth of treatment and soil suitability

Method	Depth of treatment	Soil
Vibratory roller / plate	0.5 - 2 m	Clean sand, gravel
Dynamic compaction (DC)	5 - 12 m (deeper with a heavy weight)	Granular, fill, waste, partly cohesive
Vibro-compaction	10 - 30 m	Clean sands with fines < 10 - 15 %
Compaction piles	10 - 20 m	Loose sands
Blasting	10 - 20 m	Saturated loose clean sand
Preloading, vertical drains	10 - 30 m	Soft clay, silt
Stone columns	6 - 20 m	Soft clay, silt, loose sand

2. Granular Soils: Vibration and Impact at the Ground Surface

A. Vibration at the ground surface

The surface of loose granular soil is compacted by vibrating equipment. The vibration reduces the inter-particle friction temporarily, so that the grains slide into a denser state under their own weight and the machine weight.

- Vibratory smooth drum roller (self propelled, 8 - 12 tonnes): frequency 25 - 50 Hz, amplitude 0.5 - 2 mm; 4 - 8 passes for lifts of 0.3 - 0.6 m. Best for sands and gravels.
- Vibrating plate compactor (plate vibrator) and vibratory tamper (rammer) for small areas, trenches, backfills; vibrating tyre rollers, vibrating pad foot rollers.
- Frequency should be close to the natural frequency of the soil (resonance) - sand 20 - 40 Hz; amplitude 1 - 2 mm; speed of travel 2 - 4 km/h; thickness of the layer 0.3 - 1.5 m (loose).
- Control: field density by sand replacement or core cutter, nuclear gauge; target 95 - 100 percent of the standard / modified Proctor dry density or the relative density 70 - 75 percent.
- Limitations: shallow effect (about 1 m), not for clay or silt (damping). Used for the base of the pavement, backfill, embankment, and as prior compaction of the sand fills.

B. Impact at the ground surface

- Rammers and impact rollers: heavy rammers, drop hammers and non-circular rollers (square, 3-sided impact rollers of 8 - 12 tonnes) deliver the impact energy deep (up to 2 - 3 m).
- Dynamic compaction (DC) (Menard, 1970 - France): a heavy pounder of 10 - 40 tonnes (steel or concrete block) is dropped repeatedly from 10 - 40 m height by a crane in a grid pattern on the

ground surface. Impact generates shock waves (compression, shear and Rayleigh waves) that densify the loose granular soil, collapse the voids, liquefy the soil momentarily and help to re-arrange the particles.

Depth of influence (Menard)

- $D = n \sqrt{(W H)}$ (D in m; W = weight of pounder in tonnes; H = drop height in m)
- n = 0.3 - 0.8 (0.5 typical for sand; 0.35 - 0.4 for silty, clay-fill); Example: W = 15 t, H = 20 m => $D = 0.5 \times \sqrt{300} = 8.7$ m.
- Energy per blow = $W H$; energy per unit area: 1000 to 2500 kJ/m^2 for the main pass (compare Proctor 600 kJ/m^3).

Primary grid (large spacing) - high energy → Rest for the pore pressure to dissipate → Secondary / intermediate grid → Ironing pass - low energy to compact the top

- Procedure: 2 - 3 passes; grid spacing 1.5 - 2.5 times the diameter of the pounder (4 - 10 m), 5 - 15 drops per point; the craters (1 - 2 m deep) are filled with sand or gravel after each pass; the ironing pass compacts the top 2 m using low drop.
- Waiting time: days to weeks for the pore water pressure to dissipate in the sands with fines or saturated soil (0 for the clean sand).
- Applications: loose sand, fill, landfills, collapsible soil, reclaimed land (airports - Kansai, Chennai, Mumbai) and liquefiable sites, industrial yards; rapid impact compaction (RIC) with 7 t weight, 1.2 m drop and the high frequency (40 blows/min) is a modified type.
- Precaution: ground vibration and noise (safe distance > 30 m from the structures - peak particle velocity < 5 mm/s); flying stones; the water table should be > 2 m below the surface (or a working mat of 0.5 - 1 m gravel is placed).

3. Granular Soils: Vibration at Depth

The probes or vibrators are inserted to the depth of the loose soil and densify it from within, giving uniform improvement with depth up to 30 m.

Vibro-compaction (Vibroflotation)

- Developed by Steuerman (1930s, Germany); the vibroflot is a cylinder (30 - 45 cm in diameter, 2 - 5 m long, 1.5 - 4 t) containing an eccentric weight rotated by an electric or hydraulic motor,

producing horizontal vibration (frequency 30 - 50 Hz, amplitude 5 - 12 mm). Water jets at the tip and the sides (jetting) help penetration.

- Procedure: (1) the vibrator is lowered with water jets to the design depth; (2) the vibration at the depth densifies the soil in a circle around it (radius 1.5 - 3 m) - the soil is liquefied and then it re-settles in a dense state; (3) the vibrator is lifted in stages of 0.3 - 0.5 m while the sand (backfill) is added at the top to fill the crater formed by the settlement; (4) the process is repeated at points in a triangular or square grid of 1.5 - 3 m.
- Dry bottom feed (vibro-replacement) and wet top feed methods are used.
- Suitability: clean sands and gravels with less than 10 - 15 percent fines (< 0.075 mm) and less than 2 percent clay; the grain size band of the Brown (1977) suitability number: $SN = 1.7 \sqrt{(3/D_{50}^2 + 1/D_{20}^2 + 1/D_{10}^2)}$ - excellent for SN 0 - 10, good 10 - 20, fair 20 - 30, poor 30 - 50, unsuitable > 50.
- Result: relative density increases to 70 - 85 percent; the SPT N value increases 2 - 3 times; the settlement reduced to 25 mm or less; liquefaction resistance increased. Deep treatment 10 - 20 m (up to 30 m).

Terra-probe and vibro-rods

- Terra-probe: a large steel pipe (75 cm dia, 20 m) with a vibratory hammer at the top (15 Hz) is driven (vibrated) into the sand and withdrawn at 0.3 m / 10 s; compaction in the zone of 2 - 3 m around.
- Vibro-rod / vibro-wing: a steel H-section or pipe with wings vibrated into the soil - for deep densifying of 10 - 20 m by the vibrating hammer of 15 Hz; used in the sand, cheap but a lower density increase than the vibroflot.
- Vibro-replacement / stone columns: in the sand with a high fines content the vibrator forms the hole, which is filled with crushed stone and compacted - this is the vibro-replacement (see the stone column).

Aspect	Vibro-compaction	Vibro-replacement (stone column)
Soil	Clean sand and gravel (fines < 10 %)	Silty sand, clay, soft soil
Mechanism	Densifies the existing soil	Replaces the soil with the compacted stone and reinforces
Backfill	Sand (or none)	Crushed stone 20 - 75 mm
Spacing	1.5 - 3 m	1.5 - 3 m

4. Granular Soils: Impact at Depth

Compaction piles (Compozer, sand compaction piles)

- A hollow steel casing (30 - 60 cm) is driven by the vibratory hammer into the loose soil to the depth. Sand (or gravel / crushed stone) is fed through the casing; the casing is repeatedly pulled up and re-driven (by vibration or driving) so that the sand is compacted and displaced into the soil, forming a dense sand pile of larger diameter (0.7 - 1.0 m) and densifies the surrounding soil by displacement.
- Spacing 1.5 - 2.5 m in a triangular grid; the area replacement ratio $a_s = A_{pile} / A_{tributary} = 0.1 - 0.4$ and the soil improvement is by volume displacement.
- Suitable for: loose sands, silty sand, soft clay (as a Japanese practice - sand compaction pile SCP - provides drainage + reinforcement), 10 - 30 m depth (offshore also). Not suitable when the vibration is a problem (noise, vibration to the nearby structures).
- Improvement: SPT N rises by 2 - 4 times; the bearing capacity of the composite ground improved; settlement and liquefaction reduced.

Blasting (explosive compaction)

- Explosive charges (1 - 5 kg of dynamite or TNT) are buried at the depth (5 - 20 m) in the saturated loose clean sand and detonated in the pattern. The shock wave and the sudden rise of the pore pressure cause liquefaction and then resettlement to the densified state: the settlement of 2 - 10 % of the depth.
- Several (3 - 5) explosions are repeated in a series of detonations at intervals. Cost low per m^3, quick; used for large areas, remote sites, old fill.
- Precaution: ground vibration, structures nearby; may not be used within 100 - 200 m of buildings, requires a licence and control; the effect uncertain; difficult for the controlled quality.

Deep (heavy) dynamic compaction and other impact methods

- Deep dynamic compaction with the heavy pounder 40 - 200 tonnes dropped from 25 - 40 m reaches a depth of 20 - 30 m (the Menard rule $D = n \sqrt{W H}$); also Rapid Impact Compactors, hydraulic hammer driving steel pipes / rams (Franki piles - compaction).
- Impact at depth by driven piles: displacement piles (timber, precast) installed at close spacing (compaction piles) which densify the soil between them; with the pile spacing 3 - 5 diameters.
- Vibro-driven stone columns, Deep soil mixing and jet grouting are other deep methods but they are the soil stabilisation, not the pure densification.

Method	Energy source	Depth	Remarks
Vibroflotation	Rotating eccentric (horizontal vibration)	10 - 30 m	Clean sand
Terra-probe	Vertical vibration of pipe	10 - 20 m	Cheap, slow
Compaction piles	Vibro / impact driving + sand	10 - 30 m	Displacement
Blasting	Explosives	5 - 20 m	Loose saturated sand
Dynamic compaction	Falling weight	5 - 30 m	Wide soil range

Quality control of the densification

- Before and after: SPT, CPT, DMT, shear wave velocity, plate load test; target $N_{1,60}$ > 20 - 25 (liquefaction resistant), q_c > 10 - 12 MPa, D_r 70 - 80 %.
- Check the settlement of the surface (crater volume), ground heave, lateral movements; pore pressure with piezometers to fix the rest period.

5. Cohesive Soils: Preloading or Dewatering

In the saturated clay the density increases only when the pore water drains out under a higher effective stress (consolidation). The process is slow because k is very low (10^{-7} to 10^{-9} cm/s). Preloading applies the future load (or more) before construction so that most of the primary consolidation settlement takes place beforehand.

Preloading (surcharge) method

Place the fill (surcharge) over the site → Pore water squeezed out - consolidation → Monitor settlement and pore pressure → Remove the surcharge, then construct

- A fill (earth, sand, stockpile, or water tank loaded) of height equal to or more than the structure load is placed. Surcharge (extra fill) is used to reduce the time: surcharge ratio R = Delta $sigma_s$ / Delta $sigma_f$ (typical 1.2 - 1.5 or more).
- Time of consolidation: $t = T_v H_{dr}^2 / c_v$; for U = 90 %, T_v = 0.848. The time for the soft clay of thickness 10 m (one-way drainage, $c_v = 1 \times 10^{-3}$ cm^2/s) is long: $0.848 \times (1000)^2 / 10^{-3} = 8.5 \times 10^8$ s = 27 years! Therefore vertical drains are used to shorten the time.
- Settlement: $S_c = (C_c H / (1 + e_0)) \log ((sigma_0' + Delta\ sigma) / sigma_0')$.

- With the surcharge, the soil reaches the degree of consolidation required in the shorter time under the larger load; remove the surcharge when the settlement under the surcharge equals the final settlement under the permanent load (i.e., U_{design} achieved by the Asaoka / hyperbolic method of monitoring).
- Instrumentation: settlement plates, piezometers, inclinometers, extensometers; the construction of the fill in the stages to avoid bearing failure (strength gain by consolidation; Skempton's method).

Example

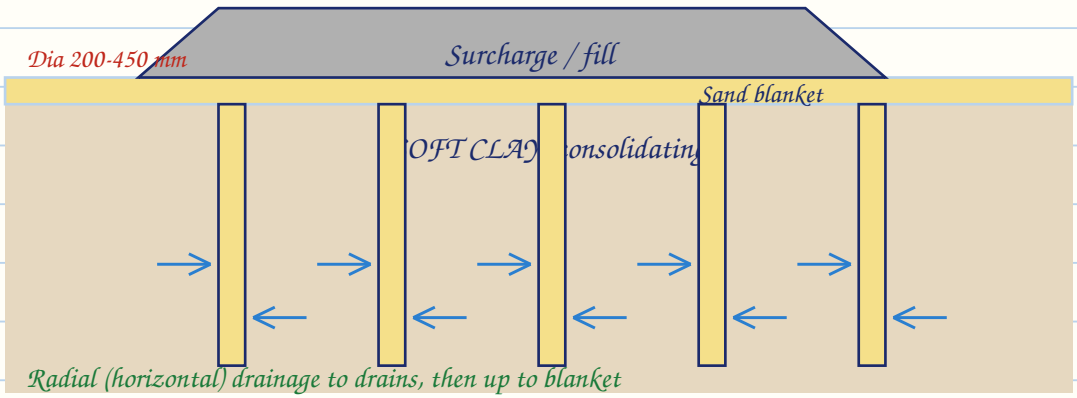
- Final settlement 0.5 m under the load p; a surcharge of 1.5 p is placed. Desired 90 % of the 0.5 m = 0.45 m.
- The surcharge causes S = 0.45 m at a U_s of 60 % of its (larger) ultimate settlement - the time required is only for U about 60 % (T_v = 0.287) instead of 90 % (T_v = 0.848) - time cut to 1/3.

Dewatering as a preload

- Lowering the ground water table by pumping (well points, deep wells) raises the effective stress by $gamma_w$ x (drop in the water table): an equivalent load of 10 kPa for each 1 m lowering. Used in the sand and silt layers and to preload by the vacuum.
- Vacuum preloading (Kjellman, 1952): an airtight membrane (PVC / LDPE film) is laid over the sand blanket, with the vertical drains connected to the peripheral trench with the vacuum pump. A vacuum of -70 to -90 kPa gives the pre-compression equal to 70 - 90 kPa of the surcharge (no risk of the bearing failure, since the confining stress does not rise), and the lateral deflection is inward. Combined with the fill: vacuum + surcharge.
- Electro-osmosis (covered in unit II) is also a dewatering method for the clay.
- Applications: highways, embankments on the soft clay (e.g., Ennore, Mumbai - Navi Mumbai airport), port and airport reclamation, storage tanks, large warehouse floors.

6. Vertical Drains - Sand Drains, Sand Wicks, Geodrains

To shorten the consolidation time, vertical drains are installed in the soft clay so that the drainage path is reduced from the layer thickness H to the horizontal radial distance (half the drain spacing, 1 - 2 m). The time of consolidation is reduced by 10 - 100 times.



Sand drains

- Vertical holes (dia 200 - 450 mm, up to 0.45 m) bored or driven in a triangular or square grid at spacing 1.5 - 4 m and filled with the clean coarse sand ($k > 10^{-2}$ cm/s, which should be at least 100 times that of the clay). Top is covered by the sand blanket (0.5 - 1 m) that carries the water away.
- Installation: displacement (driven mandrel - disturbs the clay and the smear), non-displacement (augered / jetted), the Menard and Raymond methods; the effective diameter of the drain zone d_e = 1.05 s (triangular) or 1.13 s (square).
- Limitations: smear zone, discontinuity (necking) of the sand, large settlements can cut the sand column, costly, and the sand drain needs the large quantity of sand.

Prefabricated vertical drains (PVD) - sand wicks and geodrains

- Sand wick: a sand-filled geotextile sock (small diameter 65 - 100 mm); sand drain with the filter wrap - inexpensive; better than the sand drains but higher costs and the possible bending.
- Geodrains (band drains, wick drains, PVDs): a plastic core (polypropylene, polyethylene) with grooved channels covered by a non-woven geotextile filter jacket; band 100 mm wide and 3 - 6 mm thick; equivalent diameter $d_w = (a + b) / 2$ (a = width, b = thickness), about 50 - 70 mm. Examples: Mebra drain, Colbond, Alidrain, Geodrain, Bonded fibre drain (Kjellman cardboard wick drain, 1948).
- Installation: the drain is fed through a hollow steel mandrel, pushed into the ground with the anchor plate (static push or vibratory) and cut. Speed: up to 1 m/s, 10 - 40 m deep, 1500 m of drain/h per rig. The spacing is 1 - 2 m.
- Advantages of PVD: fast, cheap, continuous, little smear and disturbance, quality assured at the factory, no sand.

Design - Barron / Hansbo radial consolidation

Design equations

- Overall degree: $(1 - U) = (1 - U_v)(1 - U_r)$ (Carrillo)
- Radial: $U_r = 1 - \exp(-8 T_r / mu)$ with $T_r = c_h t / d_e^2$
- Hansbo: $t = (d_e^2 / (8 c_h))[\ln (d_e/d_w) + (k_h/k_s) \ln s - 0.75] \ln (1 / (1 - U_r))$
- Spacing ratio $n = d_e / d_w$; $mu = (n^2/(n^2 - 1)) \ln n - (3 n^2 - 1)/(4 n^2)$ (Barron, ideal drain, no smear).
- $c_h = (2 - 5) \times c_v$ for the clay; drain spacing S chosen to give U about 90 % in the required time (3 - 6 months).

> Well resistance and smear effect ($k_s < k_h$, the disturbed zone $d_s = 2 d_m$) are accounted for; the capacity of the drain q_w should be > 100 m^3/year.

> For the depth, the drains penetrate the full compressible layer or a part, at least up to the critical depth for the stability.

7. Stone and Lime Columns

Stone columns (vibro-replacement)

> Stone columns are the vertical columns of the compacted crushed stone or gravel (dia 0.6 - 1.0 m, length 3 - 15 m, spacing 1.5 - 3 m) installed in soft clays, silts and loose silty sands by the vibroflot (vibro-replacement, wet top feed or dry bottom feed) or by the ram (Franki) or the cased auger (Rammed stone column).

> Functions: (i) reinforcement - the stiff columns carry the larger load (stress concentration ratio n = sigma$_s$ / sigma$_c$ = 3 - 6); (ii) densification of the surrounding loose soil; (iii) drainage - faster consolidation (and the dissipation of the excess pore pressure in earthquakes); (iv) reduction of settlement by 50 - 70 % and the increase in the bearing capacity two to three times; (v) improves the stability of slopes.

> Stone: 20 - 75 mm, hard, angular, durable, clean. Condition: the undrained shear strength of the clay c_u > 15 - 20 kPa (to give the lateral support to the column); for c_u < 15 kPa, the column is encased in the geosynthetic.

Design of the stone column (Priebe / Greenwood)

- *Area replacement ratio $a_s = A_c / A$ ($A = 0.866\ s^2$ for triangular, s^2 for the square grid; $A_c = pi\ d^2/4$).*
- *Ultimate capacity of the column (Hughes and Withers): $q_{ult} = sigma_{r,max}\ (1 + sin\ phi_c) / (1 - sin\ phi_c) = K_p\ sigma_{r,max}$ with $sigma_{r,max} = sigma_{r0} + 4\ c_u$ (bulging limit).*
- *Typically phi_c = 38 - 42 degree: q_{ult} about 400 - 800 kPa, safe load 100 - 400 kN per column.*
- *Priebe improvement factor $n_0 = 1 + a_s\ [\ 0.5 + f(mu_s, a_s)\ ] / (\ K_{ac}\ f(mu_s, a_s)\ ) - 1$; Settlement of improved ground $S = S_0 / n_0$.*

Lime columns and lime piles

- *Lime column (lime pile): holes (dia 0.2 - 0.5 m, depth 3 - 10 m) bored in the soft clay are filled with quick lime (CaO) or lime + sand / fly ash and compacted - used since the Roman times. Lime absorbs water from the clay (hydration: $CaO + H_2O \rightarrow Ca(OH)_2$, heat generated, volume increases 2 times), the cation exchange and the pozzolanic reaction increase the strength of the zone around.*
- *Deep lime-cement mixing (DLM / DCM): lime or lime-cement is mixed in-situ with the soil by a rotating mixing tool - the Swedish Lime-cement columns (Broms, 1970s) with dia 0.5 - 0.8 m, depth to 25 m, strength 100 - 500 kPa; used for the highway, rail embankments in the soft clay.*
- *Lime pile effect: reduction of the water content 5 - 10 %, rise in the shear strength 2 - 4 times, the soil near the pile ~ 0.3 - 0.4 m radius becomes the stabilised; improvement is mostly within 3 - 4 diameters; suitable for expansive soils (unit I) as well as soft clays and organic soil.*
- *Quantity of lime: 2 - 6 percent of the soil weight (about 10 - 20 kg per m of column).*

Aspect	Stone column	Lime column
Material	Crushed stone, gravel	Quick lime, lime-cement
Action	Reinforcement, drainage, densification	Chemical stabilisation, drying
Soil	Soft clay, silt, loose sand	Soft / expansive clay, organic
Strength gain	Immediately	With time (curing 28 days)
Settlement	Reduces by 50 - 70 %	Reduces; also controls swelling

8. Thermal Methods

Thermal methods change the soil by heating (or freezing) it to improve the strength and reduce the compressibility and volume change. They are special and costly methods used in unique situations.

1. Heating (thermal stabilisation)

- The soil is heated by burning fuel in boreholes (hot gas at 100 - 1000 C) in a series of the bores (dia 0.1 - 0.2 m, spaced 2 - 3 m) which are sealed at the top, or by the electrical heaters (conductive heating, in situ vitrification).
- Effects: at 100 C the pore water evaporates, the soil dries, the soil volume decreases; at 400 - 600 C the soil particles are irreversibly bonded (the clay minerals lose the adsorbed and lattice water and form a hard material, which no longer swells); at > 1000 C the fusion (vitrification) forms glass-like mass (brick like strength, the "fired" soil). The permeability increases (the soil is cracked), the compressibility reduces, the strength gain 10 - 100 times.
- Applications: loess and the collapsible soils, expansive clays, the soft clays around the borehole to form the stabilised soil columns (the thermal piles / "burned" columns used in Romania, Russia and Japan), and landfill / the contaminated soil (the thermal desorption of the organic pollutants).
- Limitations: very high energy cost, fuel, the effect limited to about 1 - 2 m around the hole, the safety, and the environment.

2. Ground freezing

- The soil moisture is frozen by circulating brine (calcium chloride) at -25 to -35 C or liquid nitrogen (-196 C) through the freeze pipes, which makes the ground an ice-bonded, strong and water proof wall (strength 2 - 10 MPa).
- Used as the temporary support for the deep excavations, shafts and tunnels in the saturated soil, to cut-off the water without the need of dewatering, the underpinning, and the cut-off around the contaminated sites. The freeze time is days to months; frost heave (9 % volume expansion) of the soil and the thaw settlement need the care.

Aspect	Heating	Freezing
Temperature	100 - 1000 C	-25 to -196 C
Result	Permanent improvement (hardening)	Temporary support and cut-off
Soils	Clays, loess, collapsible soils	All saturated soils
Cost	Very high	High

Remember

- Granular soil: vibration + impact (densify the grains). Cohesive soil: drain the water (preloading, drains), replace / reinforce (stone, lime columns), heat (thermal).
- Preloading + vertical drains is the most used method on the soft clay; PVD spacing 1 - 2 m, surcharge 1.2 - 1.5 times the load.

Quick revision of Unit III

- Granular: vibroflotation (clean sand, 1.5 - 3 m grid), dynamic compaction $D = n\sqrt{WH}$, compaction piles, blasting, terra-probe.
- Cohesive: preloading + surcharge, vacuum preloading (-80 kPa), sand drains (200 - 450 mm, 1.5 - 4 m), PVD (band 100 x 4 mm), stone column (stress concentration 3 - 6), lime column, thermal (400 - 1000 C), freezing.
- Hansbo / Barron design for the radial consolidation; $(1 - U) = (1 - U_v)(1 - U_r)$.

Important Questions (JNTUA pattern)

- Explain the in-situ densification methods in granular soils by vibration at the ground surface.
- Explain dynamic compaction. How is the depth of improvement estimated?
- Describe the vibro-compaction (vibroflotation) with a neat sketch. Which soils are suitable?
- Explain the compaction piles and blasting methods of densification.
- Explain the preloading method. How can the time of consolidation be reduced?
- Describe the vertical drains - sand drains, sand wick and geodrains - with their design basis.
- Explain the construction, function and design of stone columns. Compare with lime columns.
- Write short notes on thermal methods of ground improvement.

Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - IV

Soil Stabilisation: Mechanical, Cement, Lime, Bituminous and Chemical

Syllabus topics covered in this unit

- Methods of Stabilisation
- Mechanical Stabilisation
- Cement Stabilisation
- Lime Stabilisation
- Bituminous Stabilisation
- Chemical Stabilisation with Calcium Chloride, Sodium Silicate and Gypsum

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Soil Stabilisation - Introduction

Soil stabilisation is the process of *improving the engineering properties* of the soil - strength, durability, volume stability, permeability and resistance to the water and the weather - by *mechanical, chemical or physical means*, so that it can be used for the roads, foundations, embankments and the earth structures.

Objectives

- Increase the *shear strength* (CBR, UCS) and the *bearing capacity*; reduce the *compressibility and settlement*.
- Reduce the *plasticity* and *swell-shrink* (volume change) of the expansive clays.
- Reduce the *permeability* and the water absorption; improve the *durability* (freeze-thaw, wet-dry), reduce the *dust* and the erosion.
- Use the *local soil* instead of the imported material - *economy* and sustainability (reduce thickness of the pavement by 30 - 50 %).

Methods of stabilisation

Group	Methods
Mechanical	Compaction, soil blending (mixing of the different gradations), the grading, soil replacement
Chemical / admixture	Cement, lime, lime-fly ash, bitumen (asphalt), calcium chloride, sodium chloride, sodium silicate, gypsum, polymers, enzymes, resins, industrial wastes (fly ash, slag, rice husk ash, GGBS)
Physical	Heating, freezing, drainage and dewatering, electro-osmosis, grouting, stone / lime columns
Reinforcement / geosynthetic	Fibres (natural / synthetic), geotextile, geogrid, geocell, reinforced earth
Biological / new	Biocementation (MICP - Microbially Induced Calcite Precipitation), bio-polymers, enzymes

Soil type	Best stabiliser
Gravel and sand (granular, low fines)	Cement, bitumen, mechanical stabilisation
Clayey sand, silty sand, PI < 20	Cement, lime + fly ash
Clay (PI > 20), expansive soil	Lime, lime + cement, fly ash
Organic soil, peat	Cement + additives (or replace), pre-load
Dust control on gravel / unpaved roads	$CaCl_2$, NaCl, bitumen emulsion, lignosulphonate

Factors affecting the choice

* Soil type and gradation (% fines, PI), the purpose (road base, sub-grade, foundation), the climate (wet / frost), the availability and cost of the stabiliser, the construction equipment, the durability needed.

2. Mechanical Stabilisation

In mechanical stabilisation the properties of the soil are improved by changing its gradation (grain size distribution) and by compaction, without the use of the chemical admixture. It is the oldest and the cheapest method.

Principle

- A well graded soil (gravel - sand - fines) has a dense packing (the voids of the large particles filled by the smaller particles); this gives the high density, strength and low permeability, and the fine fraction supplies the cohesion (binding).
- The required gradation is obtained by blending (mixing) two or more soils (e.g., the coarse sand + the clay / the gravel + the fine soil) in the correct ratio; the plasticity should be moderate: PI 4 - 9 for the base and up to 15 for the sub-base.

Blending - the proportions

Rothfuchs / trial method

* If soils A (coarse) and B (fine) are mixed in the proportion $x : (1 - x)$ the grading of the mix at any sieve = $x\,p_A + (1 - x)\,p_B$ (p = % passing); fit it to the specification envelope (IRC / MoRTH / IS 2720).
* Example: A has 20 % passing 75 micron, B has 90 %; the target is 35 %. Then $20x + 90(1 - x) = 35 \Rightarrow x = 0.786$ (78.6 % A, 21.4 % B).

- Rothfuchs method (the triangle graph) and the graphical / trial and error methods are used for the mix design; the Proctor test gives the MDD and OMC; the CBR and the Atterberg limits check the quality.
- Typical specification of the stabilised WBM / granular sub-base (GSB): gravel 20 - 50 %, sand 30 - 60 %, fines 8 - 20 %, LL < 25 - 35, PI < 6 - 10 (IRC 37, MoRTH Clause 401 - 405).

Construction steps

Scarify and pulverise the soil → Add the blend soil and water to the OMC → Mix by the grader / pulvi-mixer → Compact by the roller (95 - 100 % MDD)

- Compaction - the static, vibratory and pneumatic tyre rollers; the layer thickness 100 - 200 mm; 95 - 98 % of the modified Proctor density; water content OMC +/- 2 %.
- Other mechanical methods: soil replacement, proof rolling, drainage (side drains, crown to give a 3 - 4 % slope), geosynthetic inclusion.

Advantages	Limitations
Cheap, simple equipment, no chemical, the local material	Strength gain is limited; sensitive to the moisture - it softens in the rain
The soil is reusable, quick	The fines should be controlled; not for the highly plastic clay or the heavy traffic on its own

3. Cement Stabilisation

Soil-cement is the mixture of pulverised soil, Portland cement and water, compacted at OMC and cured, which hardens into a rigid, durable material. Introduced in the USA in 1935; widely used for road bases, sub-bases, the canal lining, the slope protection of the dams and the foundation of the buildings.

Mechanism

- Hydration of the cement: tricalcium silicate (C_3S) and the dicalcium silicate (C_2S) react with the water to form the calcium silicate hydrate (C-S-H) gel and the calcium hydroxide $Ca(OH)_2$ - the cementing bonds between the soil particles.
- In clays the $Ca(OH)_2$ released also gives cation exchange, flocculation and the pozzolanic reaction with the clay silica / alumina - additional strength and the reduction of the plasticity.
- The strength increases with the cement content and the time (curing: 7, 28, 90 days).

Soils and the cement content

Soil (AASHTO)	Cement content by weight of the dry soil (%)
A-1-a (gravel, well graded)	3 - 5
A-1-b, A-3 (sand)	5 - 9
A-2 (silty / clayey gravel and sand)	5 - 9
A-4, A-5 (silt)	8 - 12
A-6, A-7 (clay)	10 - 15
Typical rule	Suitable for the soil with PI < 20 - 30 and fines < 35 - 50 %; the highly plastic clay and organic soil are not suited as the pulverisation is difficult

Mix design - the steps (IS 4332 / PCA method)

Classify the soil; assume the cement content → Find the OMC and the MDD of the soil-cement (IS 4332 Part 3) → Prepare the specimens (UCS), cure 7 days → Select the cement for the UCS 1.7 - 3.5 MPa; wet-dry test

- Criteria: UCS at 7 days = 1.7 - 2.5 MPa for the base, 0.7 - 1.5 MPa for the sub-base and the foundation (IRC:SP:89); freeze-thaw and wet-dry durability tests (soil-cement loss < 14 % for A-1 and A-3, < 10 % for the A-4 to A-7).
- Modified soil-cement (the less cement, 2 - 4 %) for the sub-grade modification; cement-treated base (CTB), cement-bound granular material (CBGM), the roller compacted concrete is for heavy duty.
- Construction: pulverise the soil (80 % passing the 4.75 mm sieve), spread the cement, mix (by the pulvi-mixer, the in-situ mix-in-place or the central plant mix), add water, compact within 2 hours, cure by the membrane / sprinkling / bitumen emulsion for 7 days.

Example

- Soil-cement for the sub-base: 5 % cement; the dry soil per m^3 (MDD = 1.9 t/m^3) = 1.9 t; cement = 0.05 x 1.9 = 0.095 t = 95 kg/m^3.

Advantages	Limitations
High strength and durable, a rigid layer, good load spread; reduces the thickness of the pavement	Shrinkage cracks (reflect to the surface) - use the saw cuts, the cement < 6 %
Reduces the plasticity and the swell; good for the sandy soils	Not suited for the organic soil, the sulphates and the high plastic clays; cost; the time limit for the compaction

4. Lime Stabilisation

Lime stabilisation is the addition of the *lime (quicklime CaO or the hydrated lime $Ca(OH)_2$ or the lime slurry)* to the *fine grained soil (clays)*, which reacts with the soil to reduce the plasticity, the swelling and to increase the strength. It is the best method for the *clay soils (PI > 10 - 15)*, including the *black cotton soil*.

Mechanisms (lime-soil reactions)

Reaction	Process	Effect
1. Cation exchange	Ca^{2+} replaces the Na^+, K^+, H^+ on the clay surface	Thinner double layer, *flocculation* and aggregation - reduced plasticity, easier to work with the clay (immediate, in hours)
2. Flocculation and agglomeration	Lime raises the electrolyte concentration	Soil becomes the silt-like and granular
3. Lime carbonation	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3$	Weak cementation (and loss of the lime)
4. Pozzolanic reaction	At the pH > 12.4 the *clay silica and alumina dissolve* and react with the Ca^{2+}: forms *CSH, CAH and CASH (calcium silicate / aluminate hydrates)*	Long term *strength gain* (months to years); the main strength source

- *Quicklime* reacts with the water (*slakes*), generates heat, dries the wet soils (*drying effect*, 1 % lime reduces w by about 1 %). It is used for the wet fine soil.

Quantity of the lime and the mix design

- *Lime content*: 2 - 8 percent by the dry weight (*lime modification 1 - 3 %*: the plasticity reduction; *lime stabilisation 3 - 8 %*: the strength).
- *Initial Consumption of Lime (ICL)* - the *Eades and Grim pH test*: the minimum lime content for the pH of 12.4 in the soil-lime-water mixture.
- The *UCS* target: 0.7 - 1.0 MPa at the 28 days for the sub-base (IRC: SP: 89 / IS 4332 Part 2); the CBR after the 7 days cure + 4 days soak; the plasticity reduction (PI < 10).
- *Lime-fly ash (LFA)*: lime : fly ash = 1 : 3 to 1 : 4 (the pozzolana) with the soil 10 - 20 % - slow strength gain; *lime-cement-fly ash*; *lime-sand* etc.

Construction (lime treated sub-grade)

Pulverise the soil; spread the lime → First mix; mellowing for 24 - 72 h → Remix; add the water to OMC → Compact; cure for 7 days

- The mellowing (mellowing time) lets the lime react with the clay and the lumps break; compaction by sheep-foot / pad-foot, vibratory rollers; wet curing.
- Applications: the sub-grade, sub-base and base of the roads, canal lining and banks, the foundation pads in BC soils, the lime piles and the lime slurry injection (unit I), the embankment on the soft ground, and the quick working platform.

Advantages	Limitations
Reduces the PI and the swell dramatically; good for the heavy clays; the strength rises with time; cheap; fast drying	Not suitable for the sandy / granular soils (no clay to react) and the organic soil
Improved the workability, the shrinkage reduced	Sulphate-rich soils form ettringite / thaumasite and heave - the treatment needs care; the carbonation; the dust / caustic hazard

5. Bituminous Stabilisation

Bituminous stabilisation is the mixing of the bitumen (asphalt) in the form of the cut-back, emulsion or the hot bitumen (including the foamed bitumen) into the soil. The bitumen acts as a binder and a waterproofing agent, so it is applied in the cohesionless soils (sands) and the granular soils for the flexible pavements.

Types and the action

Type	Action / soil	Binder content
Sand-bitumen (sand bitumen stabilisation)	Cohesionless sand: the bitumen coats the sand grains and binds them - gives the cohesion; for the sands, A-3, A-2 with the fines < 12 %	4 - 10 percent
Soil-bitumen (fine soil, waterproofing)	The bitumen coats the soil aggregates (lumps) and prevents the water access; keeps the moisture constant (the cohesive soil with low PI, the sand-clay)	4 - 7 percent of the cut-back / emulsion
Oiled earth (road surface)	Gravel / earth road with the sprayed bitumen on the top	Surface
Bitumen-lime / bitumen-cement	Bitumen emulsion + the 1 - 2 % cement / lime to break the emulsion in clay, the strength	-
Foamed bitumen stabilisation	The hot bitumen + 2 % cold water gives the foam (the volume 10 - 15 times), mixes with the damp, wet soil (cold recycling of the pavement)	2 - 4 percent

- Binder: cut-back bitumen (the bitumen dissolved in kerosene / the petrol - rapid (RC), medium (MC) and slow curing (SC)) for the soil-bitumen; emulsion (cationic or anionic, SS-1 / MS), tar and foamed bitumen (penetration grade 80 / 100 or VG-10 / VG-30).
- Soil suitable: the passing 75 micron < 25 %, LL < 40, PI < 18 (better for the PI < 10) in the soil-bitumen; sand-bitumen is for the dune sand - the arid deserts (Rajasthan) and the beach sand.
- Mix design: the bitumen content by the Hubbard-Field / Marshall type stability (Marshall stability > 3 kN) or the UCS and water absorption, the optimum fluid content (the water + the binder) for the maximum density; the cure.

Construction

Pulverise the soil to the fine → Add the water (pre-wetting) and the bitumen → Mix (pug-mill / in place), aerate to release the solvent → Compact; seal coat, cure

- The mixing is by the road mixer, the pug mill; the aeration to evaporate the kerosene in the cut-back; the curing time is 2 - 7 days; the surface is sealed to prevent the traffic abrasion.
- Applications: the base and the sub-base in the desert, the earth dams (the impervious facing: the asphaltic lining of the canals and reservoirs), the embankment slopes, the airfields, the shoulders, the dust control.

Advantages	Limit
Waterproof, flexible, resists the erosion and the moisture variation; works in the sand	The c high p

6. Chemical Stabilisation: Calcium Chloride

In the chemical stabilisation the small amounts of the chemical (salts, polymers, acids) change the soil-water electro-chemistry, the pore fluid, or form the cementing gel. Used for the dust control, frost resistance, the moisture control and the local strengthening.

Calcium chloride ($CaCl_2$)

- Properties: a hygroscopic salt (absorbs the moisture from the air and the humidity above 29 - 30 %), the deliquescent; a highly soluble; commercial forms flake, pellet, the brine (30 - 35 %).
- Action: (1) the moisture retention - keeps the surface damp and the soil bonded, preventing the dust and the raveling of the unpaved roads; (2) reduces the surface tension and the vapour pressure of the water, so the evaporation is reduced (the soil dries slowly) and the freezing point of the water is lowered (the frost heave is reduced - an anti-freeze); (3) the flocculation by the cation exchange (Ca^{2+}) reduces the plasticity and increases the dry density; (4) the lubricant that helps the compaction (increases the MDD with the lower OMC).
- Dosage: 0.5 - 2 % (by the dry weight of the soil) as the flakes 77 % - the 1 - 1.5 kg/m^2 for the dust control, or the 0.15 - 0.5 % in the base course.
- Applications: the unpaved roads, WBM / gravel roads, the airfield, the soil-cement curing aid (an accelerator), the cold weather concrete / the frozen soil, dust control in the mines.
- Limitations: leaches out with the rain (needs the reapplication), corrodes the metals, the harmful to the vegetation, the cost.

Remember

- $CaCl_2$: hygroscopic + deliquescent + lowers the freezing point + flocculation => used for the dust control and the frost areas. Sodium chloride (NaCl) acts similarly at the dose 1 - 3 % and is the cheaper in the arid regions.

7. Sodium Silicate and Gypsum Stabilisation

Sodium silicate (water glass, Na_2SiO_3)

- Sodium silicate (Na_2O . n SiO_2) is the viscous alkaline liquid (n = 2 - 4; the "silicate modulus"). With the reagent ($CaCl_2$, the salts, the acids, the cement, formamide) it forms the silica gel - the cementing gel (Si - O - Si network) that binds the soil particles and fills the pores.
- Reaction: $Na_2SiO_3 + CaCl_2$ -> $CaSiO_3$ (gel) + 2 NaCl; the $CaSiO_3$ / silica gel hardens, reduces the permeability and the strength of the sand rises to 0.5 - 3 MPa. The gel time (seconds to hours) is controlled by the reagent and the dilution.
- Joosten process (the 2-shot): the injection of the sodium silicate and then the $CaCl_2$ with the double tubes, the strong gel (> 3 MPa) but the lateral spread limited; the 1-shot (Rodio / Siroc): the silicate with the delayed reagent (ethyl acetate / formamide) is injected as the single liquid - the controlled gel time; the low viscosity grout (1.5 - 5 cP), the penetrates the fine sands (K > 10^{-4} cm/s).
- Applications: the underpinning, the tunnel and the excavation support in the sand, the seepage cut-off, the foundation strengthening of the old structures, the sand liquefaction prevention; the soil stabilisation in the pavement by the 2 - 5 % with the lime / cement (the alkali activation).
- Limits: the gel is less durable (the syneresis - the shrinkage, the leaching of the soda in the long term), pH high, not suited for the clays and the silt directly.

Gypsum ($CaSO_4$. 2 H_2O)

- Gypsum supplies the Ca^{2+} and the sulphate to the soil: used for the dispersive / sodic soil, the alkaline soil (the sodium rich) and the soft clay and the peat.
- Action: (1) the cation exchange (Ca^{2+} replaces the Na^+) - the flocculation, the reduction in the swell and the dispersion; (2) the slow dissolution (solubility 2 g/l) gives the long term effect; (3) with the cement / lime / fly ash, the sulphate activation helps to form the ettringite (the calcium sulpho-aluminate) that gives the early strength in the clays and the fly ash / slag blends (the gypsum activated slag).
- Dose: the 2 - 5 % (the soil-gypsum), the 5 - 10 % of the phosphogypsum / FGD gypsum (the industrial waste) in the sub-grade and the embankments, the combination 3 % cement + 3 % gypsum.
- Applications: the reclamation of the sodic (alkali) soil (the agriculture), the stabilisation of the dispersive clay in the earth dam, the soft clays (gypsum + fly ash) and the low volume roads, the gypsum plaster for the mud wall (earthen construction).
- Cautions: excessive gypsum causes the swelling by the ettringite (heave) in the stabilised soils; the sulphate attack to the concrete; the gypsum is soluble and loses the effect in the flowing water.

Chemical	Principal action	Best for
$CaCl_2$	Hygroscopic, flocculation, anti-freeze	Dust control, gravel roads, frost areas
NaCl	Moisture retention, the lowered evaporation	Arid roads (cheap)
Sodium silicate	Gel formation (silica), cementation	Sands, underpinning, cut-off
Gypsum	Ca^{2+} source, flocculation, the ettringite	Sodic / dispersive soil, soft clay
Polymers, enzymes, lignosulphonates	Bonding, the reduced water affinity	Roads, dust, special

8. Comparison and Selection of the Stabilisers

Stabiliser	Soils	Content (%)	Strength / effect	Main use
Mechanical	Any, by blending	-	CBR 20 - 80	Base, sub-base
Cement	Sand, silty and the clayey sand, PI < 20	3 - 12	UCS 1.7 - 5 MPa	Road base, the foundation
Lime	Clay, PI > 10 - 15	2 - 8	UCS 0.7 - 2 MPa, slow	Sub-grade, BC soil, canals
Bitumen	Sand, low PI soil	4 - 10	Waterproof, cohesion	Desert roads, the lining
$CaCl_2$	Gravel, the road	0.5 - 2	Dust and moisture control	Unpaved roads
Sodium silicate	Sand ($k > 10^{-4}$)	10 - 25 % volume of the grout	0.5 - 3 MPa	Underpinning, cut-off
Gypsum	Sodic / dispersive clay	2 - 10	Flocculation	Reclamation, dams

Laboratory tests for the stabilised soils

- Compaction (MDD, OMC), UCS (7, 28 days), CBR (soaked), Atterberg limits, durability (the wet - dry, freeze - thaw), free swell, the permeability, the pH (the lime); the IS 4332 (Parts 1 - 9) series - methods of test for the stabilised soils.
- Field control: the field density (the sand replacement), the plate load, DCP (Dynamic Cone Penetrometer), the Benkelman beam and the FWD; the thickness; the cement / lime content by the EDTA titration.

Environmental points and the sustainability

- The use of the industrial waste (fly ash, GGBS, the rice husk ash, the phosphogypsum, the cement kiln dust, the steel slag) as the partial stabiliser reduces the cost and the CO_2 footprint of the cement and the lime (the cement ~ 0.9 t CO_2/t).
- Possible leaching of the chemicals to the ground water (the high pH, the chlorides, the heavy metals) must be checked (the TCLP test).

Quick revision of Unit IV

- Mechanical: blending + compaction, well graded soil. Cement: the hydration, the 3 - 12 %, UCS 1.7 - 5 MPa, PI < 20.
- Lime: the cation exchange + pozzolanic, 2 - 8 %, the best for the clay / BC soil, ICL by the pH 12.4 test.
- Bitumen: sand-bitumen and soil-bitumen, 4 - 10 %, waterproofing. $CaCl_2$: hygroscopic, the dust control; sodium silicate: the silica gel, the Joosten 2-shot; gypsum: Ca^{2+} source, the sodic soil.

Important Questions (JNTUA pattern)

* Explain the different methods of soil stabilisation and the factors governing the choice of the method.
* Describe the mechanical stabilisation of the soil. How is the proportion of the blending determined?
* Explain the mechanism and construction of the cement stabilised soil. Discuss the cement content for the different soils.
* Explain lime stabilisation. Discuss the lime - soil reactions and the mix design (ICL).
* Describe the bituminous stabilisation of the soil with the types of the binders.
* Explain the chemical stabilisation with $CaCl_2$, sodium silicate and gypsum.
* Compare the cement, lime and bitumen stabilisation. Which one is suited for the expansive soil?

Subject Code : 23A01703a

GROUND IMPROVEMENT TECHNIQUES

UNIT - V

Reinforced Earth and Geosynthetics

Syllabus topics covered in this unit

- Reinforced Earth: Principles
- Components of Reinforced Earth
- Factors Governing Design of Reinforced Earth Walls
- Design Principles of Reinforced Earth Walls
- Geosynthetics: Geotextiles - Types, Functions and Applications
- Geogrids and Geomembranes - Functions and Applications

How to use these notes

- Red words = keywords to remember for the exam.
- Green words = examples, units and values.
- Yellow boxes = quick points; pink boxes = important questions / formulae.
- Learn the sketches - they fetch marks in the answer paper.

1. Reinforced Earth - Principles

Reinforced earth (reinforced soil) is a composite material formed by the soil (backfill) and the tension-resisting reinforcing elements (strips, grids, sheets, fibres) placed in the soil in layers. It was invented and patented by the French architect-engineer Henri Vidal in 1963 (first wall built in 1965 at Pragnères, France) and named "Terre Armée". In India it is now widely used for flyover approaches, metro viaducts and highway embankments.

Basic principle

- The soil is strong in compression and shear but has no tensile strength. The reinforcement is strong in tension. When the soil mass tries to deform laterally under the load, the friction (or the bearing / the interlocking) between the soil and the reinforcement mobilises the tension in the reinforcement, which restrains the soil - the composite behaves as if it has the cohesion (the apparent cohesion) - "The reinforcement carries the tension and the soil carries the compression".
- The soil-reinforcement interaction is by friction (strips, sheets) and passive resistance / interlocking (grids, anchors).
- The facing (skin) holds the soil at the face and prevents the local sloughing between the reinforcement layers; the real function is that of the soil retention, not the earth pressure resistance - the reinforced mass acts as a gravity wall.

Two views of reinforcement action

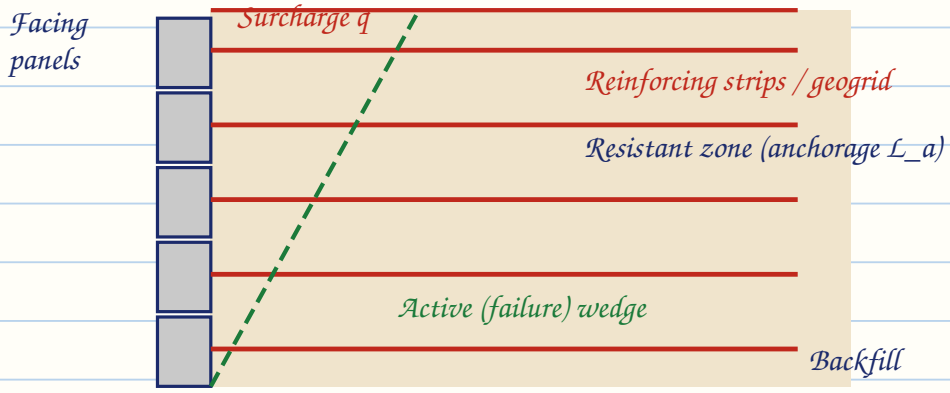
Concept	Explanation
Internal confinement (Vidal) - Apparent cohesion (Schlosser and Long)	The reinforcement restrains the lateral strain of the soil; the soil mass has the anisotropic apparent cohesion: $c_r = T_R \, \mathrm{sqrt}(K_p) / (2 S_v S_h)$, where T_R = the tensile strength of the reinforcement and S_v, S_h are the spacing
Enhanced confining stress (Mohr circle)	The reinforced soil shows the higher strength since the reinforcement raises the minor principal stress: $\mathrm{sigma}_1 = \mathrm{sigma}_3 K_p + 2 c_r \, \mathrm{sqrt}(K_p)$; the failure envelope shifts upward

Advantages of reinforced earth

- Flexible, tolerates the large differential settlement; the quick construction (precast facing); the economical for the height > 4 - 6 m (30 - 50 % saving on the RC walls); the ductile - excellent in the earthquakes (performed well in the Kobe, Northridge); the aesthetics (the precast panels); the less foundation preparation - the walls up to the 20 - 30 m high.
- Limitations: needs the good granular backfill and the space (the reinforcement length 0.7 H); the durability of steel in the aggressive soil; the specialised design and the drainage care; the not suited to the walls in the water (the scour).

Applications

- Retaining walls at the bridge abutments, the approach embankments, the flyovers and the metro; the steep slopes (the reinforced soil slopes - RSS, 1 H : 2 V to 1H : 1V); the embankments on the soft soil (the basal reinforcement); the foundations (the reinforced soil footing - the bearing capacity); the pavement sub-base, the landslide repair, the dams / the dykes, the mining, the noise barrier mounds, the anchors.



2. Components of Reinforced Earth

A reinforced earth wall consists of four main components: the backfill soil, the reinforcement, the facing and the foundation + drainage.

1. Backfill (the fill soil)

- The frictional (cohesionless), the free draining granular soil is the best: the sand, gravel, crushed rock; the angle of friction phi >= 30 degrees (the typical 34 - 36); the fines (< 75 micron) not more than 15 % (FHWA: < 15 %; the PI < 6, the clay content < 10 %); the maximum size 75 - 100

mm (but the 19 mm to avoid the damage of the geosynthetic); the coefficient of the uniformity C_u > 2 - 4.

- Chemical limits for the steel reinforcement: pH 5 - 10, the resistivity > 3000 ohm-cm, the chlorides < 100 ppm, the sulphates < 200 ppm (the organic content < 1 %); the soil with the low pH 4.5 - 9 for the geosynthetics (the polyester pH 3 - 9, the polypropylene / polyethylene are the more resistant).
- The fine-grained and the cohesive soils are used with the geosynthetic (a drainage geocomposite) with the care (the pore pressure).

2. Reinforcement

Type	Description	Features
Metallic strips	Galvanised steel strips - ribbed, 50 - 120 mm wide x 3 - 5 mm thick (typical 50 x 4 mm), the grade Fe 410 / 500; stainless steel and aluminium for special	High tensile strength (up to 400 MPa), the stiff, low strain; the corrosion needs the sacrificial thickness (galvanising 0.086 mm, the loss 0.015 mm / year first 2 years, then 0.004 mm/yr)
Metallic grids / bar mats / welded wire mesh	The steel bar mat (8 - 10 mm bars), the wire mesh	Passive resistance via the cross bars (the more pullout resistance)
Geogrids (polymeric)	The HDPE / polyester (PET) / PP grids, the uniaxial	The non-corroding, the flexible, the interlocking; creep (reduced strength factor)
Geotextiles (woven)	The woven polyester or PP sheets (strength 20 - 400 kN/m)	The low cost; the friction; used for the slopes
Geocells, fibres (the randomly distributed), the soil nails and anchors	The honeycomb, the discrete fibres in the soil (the fibre-reinforced soil), the nails	For the local reinforcement
Natural (jute, coir, bamboo)	Geo-jute, coir geotextile	The temporary / low cost / biodegradable

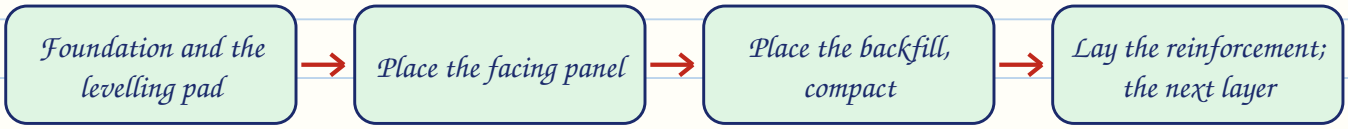
3. Facing (skin element)

- Precast concrete panels (cruciform, 1.5 x 1.5 m, thickness 140 - 180 mm, the hexagonal, the rectangular, the segmental block (interlocking dry-stack blocks), the galvanised steel half-pipe / the sheets (the corrugated, the older type), the welded wire mesh facing (the wrapped geosynthetic - the wrap-around facing), the gabion facing, the shotcrete, the vegetated (green) facing.

- The facing is flexible - the joints with the filter / the geotextile to prevent the loss of the soil (and the neoprene / the polyethylene bearing pads to allow the movement); the facing is connected to the reinforcement by the bolts, the clips, the loops.

4. Foundation and drainage

- A levelling pad (the unreinforced concrete, 150 mm x 300 mm) at the base of the facing; the foundation soil should be able to support the pressure (the bearing, the settlement - the soft soil is improved); the embedment depth = 0.1 H or min 0.5 m (the 1 m).
- Drainage: the drain behind the facing, the weep holes, the geocomposite drain, the blanket drain at the base and the interceptor ditches on the top; the good drainage is essential - no hydrostatic pressure should build up.



3. Factors Governing the Design of Reinforced Earth Walls

Factor	Influence on the design
Wall geometry	The height H, the face batter (the vertical / the inclined), the length of the reinforcement (L / H = 0.7 or more, min 2.5 - 3 m), the top slope (the sloping backfill, the surcharge)
Backfill soil properties	The unit weight gamma, the angle of friction phi (the strength, which determines the K_a, the pullout), the gradation, the drainage, the compaction (95 % MDD), the chemical (corrosion)
Reinforcement properties	The tensile strength T_{allow}, the stiffness (extensible / inextensible), the soil-reinforcement friction coefficient f* or the interface angle delta, the creep, the installation damage, durability and the chemical degradation, the width and the spacing (S_v, S_h), the length L
Loads	The dead loads (the soil weight), the surcharge (the traffic 12 - 20 kPa, the equipment, the buildings), the seismic (the horizontal acceleration k_h), the water, the impact / vehicle collision on the barriers (the parapet)
Foundation soil	The bearing capacity, the settlement (the total and the differential), the slope stability, the ground water; the improvement (the replacement, the stone columns, etc.)
Facing	The type, the connection strength, the flexibility (allowable differential settlement = 1 in 100), the aesthetics
Drainage and the environment	The ground and the surface water, the seepage, the flooding and scour at the toe (the wall in the water), the freeze - thaw
Construction	The compaction near the facing (the light, the 1 m zone), the quality, the construction tolerance, the service life (75 - 120 years for permanent walls: the corrosion allowance), the cost, the availability of the material

- Safety factors (BS 8006 / FHWA / AASHTO / IS 14458 (Part 1 - 3) for the retaining walls in India): the FS against sliding >= 1.5, overturning >= 2.0, the bearing >= 2.5, the pullout >= 1.5 (the reinforcement tension FS 1.5 - 2.5 for the metallic) and the global stability >= 1.3 - 1.5.
- Extensible reinforcement (geosynthetics, a larger strain) vs the inextensible (the steel - the strain less than the soil): they follow the different design approaches (the K-stiffness method and the coherent gravity method).

4. Design Principles of Reinforced Earth Walls

The design of a reinforced earth wall has two parts: the external stability (the whole reinforced mass as a gravity retaining wall) and the internal stability (the reinforcement does not break or pull out). The soil pressure is Rankine active in the retained backfill.

A. External stability (the reinforced block of the width L and the height H)

Checks (the Rankine, coefficient $K_a = \tan^2(45 - \phi/2)$)

- Active thrust of the retained fill: $P_a = 0.5\, K_a \gamma_b H^2$ (+ $K_a q H$ for the surcharge q), acts at H/3.
- FS sliding = $(W \tan \delta_b + c L) / P_a >= 1.5$, $W = \gamma_r H L$ (the weight of the reinforced block); the base friction $\delta_b = \min(\phi_r, \phi_f)$ (the foundation).
- FS overturning = $(W L / 2) / (P_a H / 3) >= 2.0$ (the resultant within the middle third: the eccentricity $e = L/2 - (M_R - M_O)/W < L / 6$ - for the rock L/4).
- Bearing pressure at the base (Meyerhof): $\sigma_v = W / (L - 2e) <= q_{ult} / 2.5$ (the rigid foundation) ; the settlement is checked. Global slip circle analysis (the Bishop) also to be done.

B. Internal stability - the tension and the pull-out (the Rankine method / the Coherent gravity method)

- Lateral earth pressure at the depth z in the reinforced soil: $\sigma_h = K \gamma z$ (+ Kq), $K = K_a$ for the extensible (the geosynthetic - the K constant with depth), K varies from the $K_0 = 1 - \sin\phi$ at the top (z = 0) to K_a at the 6 m depth for the inextensible (steel strips) - the coherent gravity (Juran and Schlosser) method.
- Tension in the reinforcement at the depth z with the vertical spacing S_v and the horizontal spacing S_h: $T = \sigma_h S_v S_h$ (per strip). For the continuous sheets (the per m width): $T = \sigma_h S_v$.
- The failure surface: the Rankine plane at 45 + phi/2 from the horizontal passing through the base of the facing (the inextensible: the maximum tension line - the curved, at 0.3 H from the face at the top to 0.5 (H - z) tan(45 - phi/2) at the base).

Design of the reinforcement

- Tension failure: $T_{max} <= T_{allow} = f_y A_c / FS$ (the net area $A_c = w$ (t - corrosion loss)), $FS_T = T_{allow} / T >= 1.5$ (the steel: 0.55 f_y). For the geosynthetic: $T_{allow} = T_{ult} / (RF_{ID} \times RF_{CR} \times RF_D \times FS)$ with the reduction factors for the installation damage, the creep and the durability.
- Pull-out (the anchorage length L_e) beyond the failure surface: $P_R = 2\ b\ L_e\ sigma_v\ f^*$ (for the strip, the two faces, b = width, f^* = the apparent friction coefficient = tan delta, 1.5 at the top to tan phi at the 6 m depth for the ribbed steel strips; 0.6 - 0.8 tan phi for the geotextile)
- $FS_{pullout} = 2\ b\ L_e\ gamma\ z\ f^* / T >= 1.5 => L_e = FS \times T / (2\ b\ gamma\ z\ f^*)$; Total length $L = L_e + L_r$, where L_r = the length in the active wedge = (H - z) tan(45 - phi/2) (the Rankine).
- Spacing: $S_v = T_{allow} / (sigma_h\ S_h)$ (the typical S_v = 0.3 - 0.75 m for the steel / 0.3 - 0.6 m for the geogrid, the S_h = 0.5 - 1.5 m; the spacing is smaller with the depth).

Fix H, backfill (gamma, phi), the surcharge → Check the external stability (L = 0.7 H) → The tension and the pull-out at each layer → Facing connection, drainage, the global check

Example (the geogrid wall, per m run)

- H = 6 m, gamma = 18 kN/m^3, phi = 32 degree, K_a = 0.307, no surcharge, the S_v = 0.6 m, the geogrid T_{allow} = 30 kN/m.
- At the base layer (z = 5.7 m): $sigma_h = 0.307 \times 18 \times 5.7$ = 31.5 kPa; $T = 31.5 \times 0.6$ = 18.9 kN/m < 30 kN/m (FS = 1.59 > 1.5 OK).
- The anchorage with the tan delta = 0.67 tan 32 = 0.42: $L_e = FS\ T / (2\ gamma\ z\ tan\ delta)$ = 1.5 x 18.9 / (2 x 18 x 5.7 x 0.42) = 0.33 m, therefore the minimum 1.0 m governs; L_r = (6 - 5.7) tan(29) = 0.17 m; top layer L_r = 5.7 x 0.554 = 3.2 m; hence L = 0.7 H = 4.2 m is adopted.

Seismic design

- Mononobe-Okabe the pseudo-static: the dynamic increment $P_{AE} = 0.5\ gamma\ H^2\ (K_{AE})$ and the inertia force of the reinforced block (the 50 % of the dynamic thrust acts at the 0.6 H); the additional tension; the reinforcement length and the pull-out are checked for $k_h = 0.5\ A_{max}$ (the

IS 1893).

5. Geosynthetics - Introduction and Geotextiles

Geosynthetics are the planar, synthetic (polymeric) products - made of the polypropylene (PP), polyester (PET), polyethylene (PE / HDPE), polyvinyl chloride (PVC), the nylon, the polyamide - used with the soil, rock or the other geotechnical material as an integral part of the civil engineering structure. Their use started in the 1960s (the first woven geotextile in the 1958, Florida - the erosion control; Rhone Poulenc 1970s - the non-woven).

Functions of geosynthetics (SFDRC)



Geotextile: S F D R | Geogrid: R | Geomembrane: C (barrier)

Types of the geosynthetics

- Geotextiles (the woven, the non-woven, the knitted), Geogrids, Geomembranes, Geonets, Geosynthetic clay liners (GCL), Geocomposites (the drain), Geocells, Geopipes, Geofoam (EPS blocks), the geo-strips.

Geotextiles - types

Type	Manufacture	Properties
Woven	The weaving of the slit films / the monofilament yarns on a loom (the warp and the weft)	The high strength and low strain (20 - 800 kN/m); low permeability (the less open area); the reinforcement and the separation
Non-woven needle-punched	The fibres (staple or the continuous) are mechanically bonded by the needles	The thick, the high permeability and the flexible, the good in-plane drainage, the filtration, the cushion (the protection), the strength 8 - 30 kN/m, the elongation 50 - 100 %
Non-woven heat-bonded / spun-bonded	The fibres are bonded by the heat (the thermal) or the resin	The thin, the stiff, the filtration
Knitted	The interlocking loops	Special - the high strain
Composite / the geotextile-geogrid	The woven + non-woven	The strength + the filter

Functions of the geotextiles (the 5 functions - SFDRP)

Function	Description	Application
Separation	Keeps the two dissimilar soils (the soft sub-grade and the aggregate base) from mixing; keeps the base thickness	The unpaved and the paved roads, the railways (the ballast), the airfields, the parking
Filtration	Allows the water to pass and retains the soil particles (a replacement of the sand filter)	The drains (the French drain, the blanket / the toe drains), behind the retaining walls, the riprap and the revetments, the dams
Drainage (in-plane)	The transmission of the fluid in the plane of the thick non-woven	The drainage of the pavement, the retaining walls, the landfill leachate
Reinforcement	The tensile strength increases the bearing capacity, the embankment stability	The soft ground embankments, the steep slopes, the walls, the sub-base
Protection (cushion)	The cushioning of the geomembrane against the puncture	Below / above the geomembrane in the landfill, the ponds
Erosion control	Prevents the soil loss on the slope	The coastal, the river bank protection, the silt fence

Properties and tests of the geotextiles

- Physical: the mass per unit area (the g/m^2, 100 - 800), the thickness (IS 13162 Part 3), the apparent opening size (AOS, O_{95}); Mechanical: the wide-width tensile test (IS 13162 Part 5, the

strip 200 mm), the CBR puncture, the trapezoidal tear, the burst; Hydraulic: the permittivity (psi = k_n / t, 1/s), the in-plane transmissivity, the gradient ratio / the clogging; Endurance: the UV resistance, the creep, the chemical and the biological resistance.

- Filter design (the soil retention): O_{95} <= B D_{85} (B = 1 for the coarse; the 0.5 - 1.8) and the permeability $k_{geotextile}$ >= k_{soil}; the clogging (the porosity > 30 %, the AOS > 3 D_{15}).

Applications of the geotextiles

- Roads: the separation + the reinforcement on the soft sub-grade (the reduction of the aggregate thickness 30 - 50 %); the railway track; the airport runway; the asphalt overlay (the paving fabric to stop the reflection cracking).
- Drainage: the filter wrap of the sub-surface drains, the toe filter of the dams, the vertical drain jacket; the reinforced soil (the walls, the slopes, the embankment on the soft clay); the erosion control (the river training, the coastal revetments, the geotube (the sand-filled big tubes for the breakwater and the land reclamation)); the silt fences.

6. Geogrids - Functions and Applications

Geogrids are the open grid-like polymeric structures with the large apertures (the openings 10 - 100 mm) and the tensile ribs (the longitudinal and the transverse) which interlock with the soil / the aggregate particles; their main function is the reinforcement. The geogrid-soil interaction is the friction on the ribs and the passive bearing resistance on the transverse members.

Type	Manufacture	Features
Uniaxial (the tensile strength in one direction)	The extruded HDPE sheet punched and stretched in one direction; or the woven / the knitted PET coated with the PVC / the bitumen; the polyester yarn grids	The strength 40 - 400 kN/m; the low creep; the retaining walls, the steep slopes, the embankments on the soft soil
Biaxial (the same in both)	The punched and stretched in both directions; the PP	The strength 20 - 60 kN/m in both; the road base, the railway sub-ballast, the working platform on the soft ground
Triaxial / Multi-axial	The triangular apertures	The radial stiffness, the good confinement of the aggregate
Welded / the bonded (the flexible)	The polymer straps welded at the junctions	The high junction strength

Functions of the geogrids

- Reinforcement (the main): the tensile strength, the lateral restraint (the confinement of the aggregate that increases the stiffness, the "mechanical stabilisation" and the load spreading in the base layer), the bearing capacity improvement (the tensioned membrane effect).
- Separation and the stabilisation (the minor, the interlock with the aggregate), the strengthening of the asphalt (the grid in the overlay, the cracking control).

Applications

- Reinforced earth walls and steep slopes (with the wrap-around or the modular block facing); the embankment over the soft clay (the basal reinforcement; the stability and the reduced settlement); the bridge abutment, the landslide repair.
- Unpaved / paved roads over the soft sub-grade (the base reinforcement: the thickness reduced up to 40 %, the traffic benefit ratio TBR of 2 - 10 times, the rutting reduced), the railway ballast, the airfield; the foundation on the weak soil (the reinforced sand layer, the bearing capacity ratio BCR 1.5 - 3); the piled embankments (the load transfer platform); the landfill capping and the void bridging (the sinkholes, the mines).
- Geocells (the 3-D honeycomb grids) for the load support, the slope and the channel protection, the geo-webs.

Design strength of the geogrid

- $T_{allow} = T_{ult} / (RF_{CR} \times RF_{ID} \times RF_{D} \times FS)$ (the creep 1.5 - 2.5 for the PP, 1.3 - 1.6 for the PET; the installation damage 1.05 - 1.4; the durability 1.1 - 1.3; FS = 1.3 - 1.5 for the overall uncertainty).
- Example: T_{ult} = 100 kN/m; RF_{CR} = 1.6, RF_{ID} = 1.1, RF_{D} = 1.1, FS = 1.3 => T_{allow} = 100 / (1.6 x 1.1 x 1.1 x 1.3) = 39.7 kN/m.

7. Geomembranes - Functions and Applications

Geomembranes are the continuous, thin, flexible, practically impermeable polymeric sheets (the permeability 10^{-11} to 10^{-13} cm/s; the thickness 0.25 - 3 mm, the rolls 7 - 10 m wide). The main function is the barrier (the containment / the liquid and the vapour barrier).

Material	Features	Typical thickness
HDPE (high density polyethylene)	The most widely used: the high chemical resistance, the strong, the UV resistant (the carbon black), the stiff, the weldable by the thermal fusion / the extrusion; but the stress cracking	1.0 - 2.5 mm
LLDPE / LDPE	More flexible, the better conformability to the deformation	0.5 - 1.5 mm
PVC (polyvinyl chloride)	The flexible, the easy to install, the solvent / the thermal welded; the plasticiser lost (the ageing)	0.5 - 1.5 mm
fPP (flexible polypropylene), EPDM rubber, CSPE (Hypalon), the bituminous geomembrane	The special: the roofing, the tanks, the canals	1 - 2 mm

Functions

- Barrier / the liquid containment (the main): to prevent the flow of the liquid or the gas (the seepage, the leachate, the contaminants); the vapour barrier; the moisture control (the expansive soil - the vertical and the horizontal moisture barrier, the unit I).
- The protection of the soil from the water / the chemical; the separation (the minor).

Applications

- Landfills: the liner and the cover (the composite liner = HDPE geomembrane (1.5 - 2 mm) + the compacted clay liner / the GCL with the leachate collection above); the hazardous waste containment, the mine tailings and the heap leach pads.
- Canals, reservoirs and tanks: the lining to reduce the seepage losses (the irrigation canals - the water saved 20 - 40 %), the farm ponds, the fish ponds, the dams (the upstream face, the core), the tunnel water proofing, the roof gardens.
- Underground storage tanks (the secondary containment), the oil / the chemical storage, the sewage lagoons, the moisture barrier under the pavement and the foundation in the expansive soil, the floating covers (the biogas).
- Installation: the sub-grade preparation, the anchor trench, the seaming (the wedge / the extrusion welding with the quality control by the air pressure and the vacuum box tests), the protection by the non-woven geotextile (the 400 - 800 g/m^2) cushion, the ballast; the wrinkles and the puncture are the major risks; the leak detection (the electrical, the spark test).

Other geosynthetics

- Geonets (the drainage core, the 5 - 10 mm thick, the crossing ribs - the in-plane drainage in the landfill leachate collection); GCL (the bentonite sandwiched between the geotextiles, the $k < 10^{-9}$ cm/s, the self-sealing); geocomposites (the geotextile + the geonet / the grid; the prefabricated vertical drains); the geopipes (the perforated, the corrugated).

Geosynthetic	Main functions	Typical application
Geotextile	S F D R P	Roads, the filter, the drains
Geogrid	R (the confinement)	Walls, the soft soil, the road base
Geomembrane	Barrier	Landfill, the canal and the pond lining
Geonet / composite	D (drainage)	Leachate collection, the sports field
GCL	Barrier	Landfill cap, the liner

Quick revision of Unit V

- RE = soil + the tension elements; the friction; the apparent cohesion; the Vidal 1963. Components: the backfill (phi >= 30, the fines < 15 %), the reinforcement (the steel strips, the geogrid), the facing, the drainage.
- Design: external (the sliding 1.5, the overturning 2, the bearing 2.5), the internal: $T = K \gamma z S_v S_h$, the pull-out $L_e = FS\ T / (2\ b\ \gamma\ z\ f^*)$, $L = 0.7\ H$.
- Geotextile: separation, filtration, drainage, reinforcement, protection. Geogrid: the reinforcement by the interlock. Geomembrane: the barrier (HDPE) for the landfill and the canal lining.

Important Questions (JNTUA pattern)

* Explain the principle of reinforced earth. List the advantages and applications.
* Describe the components of a reinforced earth wall. What are the requirements of the backfill?
* Discuss the factors governing the design of reinforced earth walls.
* Explain the design procedure of a reinforced earth wall - the external and the internal stability. A 6 m high wall with phi = 30 degree and gamma = 18 kN/m^3 is reinforced with the strips at S_v = 0.75 m; find the tension at the base layer.
* What are geotextiles? Explain their types, functions and applications.
* Explain the types, functions and applications of geogrids.
* Describe the geomembranes: the materials, the functions and the applications in landfills and canals.